Zero-Knowledge Proofs of Matrix Multiplication from Lattice Assumptions:
Theory and a Complete Implementation Blueprint

Working technical report

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Abstract

We give a complete, self-contained treatment of two zero-knowledge proof protocols for the statement $C = AB$ over matrices: protocol Π_1, in which A is public and B, C are hidden, and protocol Π_2, in which all three matrices are hidden. Both protocols follow a *commit–compress–mask* design: the hidden matrices are pinned down by lattice-based homomorphic commitments (binding under the Short Integer Solution assumption, hiding either statistically via the leftover hash lemma or computationally under a knapsack-LWE assumption), the matrix claim is compressed to a vector or scalar claim by Freivalds-style random projection, and the compressed claim is proven by a masked Σ-protocol with rejection sampling. All operations are integer linear algebra modulo a word-sized prime; no elliptic curves or pairings are used. Part I develops the theory from first principles, assuming no cryptographic background: definitions, all supporting lemmas with proofs, the two protocols with completeness, knowledge-soundness, and zero-knowledge arguments, an analysis of why computational binding is unavoidable, and a concrete-hardness analysis of the cost of every attack. Part II is an implementation blueprint: a full symbolic parameter-constraint system with worked illustrative parameter sets, complete pseudocode for every algorithm including Fiat–Shamir transcripting and rejection sampling, arithmetic and performance engineering, cost accounting, an implementation-security checklist, and a testing and validation plan. The protocols are matrix-multiplication specializations of the BDLOP/ALS family of lattice proof systems; we make no novelty claims and give attributions throughout.

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Part I
Theory

1 Introduction

1.1 The problem

A prover $\mathcal{P}$ has computed a matrix product. A verifier $\mathcal{V}$ wants assurance that the product is correct, but some of the matrices involved are proprietary and must remain hidden. We treat two settings.

Setting 1 (public A). Matrices $B \in \mathbb{Z}_q^{n\times m}$ and $C = AB \in \mathbb{Z}_q^{n\times m}$ are known only to $\mathcal{P}$; the matrix $A \in \mathbb{Z}_q^{n\times n}$ is public. This models, for instance, proving that a public linear operator was applied to a secret input, or — transposing roles — verified inference against committed proprietary model weights.

Setting 2 (all hidden). A, B, and $C = AB$ are all known only to $\mathcal{P}$. The verifier holds only *commitments* to the three matrices (sealed, tamper-evident containers defined in Section 4) and must be convinced that the sealed values satisfy the multiplicative relation.

In both settings we demand the three standard guarantees, made precise in Section 2: *completeness* (an honest prover convinces the verifier), *knowledge soundness* (a cheating prover, whose matrices do *not* satisfy $C = AB$, convinces the verifier only with negligible probability — and in fact any convincing prover demonstrably "knows" satisfying matrices), and *zero-knowledge* (the verifier learns nothing about the hidden matrices beyond the truth of the statement).

1.2 Design goals and the shape of the solution

The motivating engineering constraint is to avoid the expensive cryptographic operations (elliptic-curve multi-scalar multiplications, pairings, FFTs over large prime fields) that dominate the runtime of general-purpose proof systems, and instead to build the entire protocol out of *integer linear algebra modulo a word-sized prime* — the same kind of arithmetic as the matrix multiplication being proven. The resulting design rests on three ingredients.

1. **Freivalds compression** (Section 3). Probabilistic verification of matrix products: if $(C - AB)\mathbf{v} = 0$ for a random vector $\mathbf{v}$, then $C = AB$ with high probability. This collapses an $n \times m$-sized claim to an n-sized (or, with two-sided projection, scalar-sized) claim, and is the source of the protocol's efficiency. It is unconditionally sound — no cryptography — *when the matrices are public.*

2. **Homomorphic lattice commitments** (Sections 4 and 5). Because the matrices are hidden, something must pin down *which* matrices the proof is about, and must do so *before* the random projection vector is revealed (otherwise the prover adapts). Commitments provide exactly this temporal binding. Ours are built from random integer matrices modulo q; their binding rests on the Short Integer Solution (SIS) problem — the precise form of the intuition that it is hard to find short integer vectors in the kernel of a random matrix — and their hiding is available in two modes, statistical (leftover hash lemma) or computational (knapsack-LWE), with a quantified cost difference.

3. **Masked Σ-protocols with rejection sampling** (Section 6). The compressed claim is proven by a three-move challenge–response protocol in which the prover reveals only *noise-masked* combinations of its secrets. Uniform masking modulo q handles the message-valued parts perfectly; Gaussian masking with Lyubashevsky-style rejection sampling handles the parts that must remain short. Fresh noise is drawn for every challenge; a mask is a one-time pad and absorbs exactly one revelation.

Protocol Π_1 (Section 7) combines these for the public-A setting, where the relation $C = AB$ is *linear* in the secrets. Protocol Π_2 (Section 8) handles the all-hidden setting, where the relation is *bilinear*: a second Freivalds projection compresses the claim to a single inner product of two committed vectors, and a classical quadratic "garbage-term" Σ-protocol handles the one remaining multiplication of secrets. Crossing the linear/bilinear boundary is the entire difference between the two protocols, and it is crossed at its narrowest point: exactly one product gate.

1.3 Attribution and lineage

Nothing in this document is new cryptography. The commitment scheme is the Ajtai-style/BDLOP construction [6, 15]; the masking discipline is Lyubashevsky's Fiat–Shamir-with-aborts [10, 11]; the product Σ-protocol of Π_2 is the lattice adaptation of a classical technique, developed in [18]; Freivalds' check is from 1977 [1]. What this document contributes is a specialization of that machinery to the matrix-multiplication relation, worked out end to end at implementation depth, with every proof and every parameter constraint spelled out. Readers wanting the state of the art in general lattice proof systems should consult [21, 22, 23].

1.4 Relation to the informal design sketch

This report grew out of an informal protocol sketch, and it is worth recording precisely how the rigorous version refines it, because each refinement is forced by an identifiable mathematical constraint.

1. *Small-coefficient challenges and projections.* The informal sketch drew the Freivalds vector $\mathbf{v}$ uniformly from $\mathbb{Z}_q^m$ and let challenges be arbitrary scalars. The rigorous version draws the entries of $\mathbf{v}$ (and $\mathbf{s}$, and the challenge γ) from a *small* set $\mathcal{S}_b = \{0, \ldots, 2^b - 1\}$ with $2^b \ll q$. The reason is norm bookkeeping: commitment openings must remain *short* for binding to reduce to SIS, and homomorphically combining commitments multiplies opening norms by the combination coefficients. Uniform mod-q coefficients would destroy shortness, and with it the honest prover's ability to respond. Smaller per-challenge sets are compensated by parallel repetition (Section 7.5); the trade-off is quantified in Part II.

2. *Uniform masking in the message slot.* The informal sketch masked with large bounded noise ("ε large enough to obfuscate B"). In the rigorous version the masked quantities split into two kinds. Message-valued responses (such as $\mathbf{z} = \boldsymbol{\varepsilon} + \gamma\mathbf{b}$) are masked with noise *uniform modulo* q, which hides perfectly — the compactness of $\mathbb{Z}_q$ dissolves the "how large is large enough" question entirely. Only the short randomness-valued responses need the careful Gaussian rejection-sampling treatment of Section 6.

3. *Two hiding modes, stated honestly.* The accompanying discussion asserted that recovery of B is information-theoretically impossible. That claim is correct for the *statistical* hiding mode (leftover hash lemma), whose parameter surcharge we quantify in Section 5.4 and Part II; the

efficient instantiation uses computational hiding under knapsack-LWE, for which secrecy of B is a 2^{128}-hardness statement of the same lattice family as binding, rather than an unconditional one. Section 4.2 explains why one of the two guarantees must be computational, and the deployment chooses which.

1.5 Roadmap

Part I: Section 2 fixes definitions; Sections 3–6 develop the three ingredients with proofs; Sections 7–8 give the protocols and their security theorems; Section 9 analyzes concrete attack costs. Part II: Section 10 chooses instantiation routes; Section 11 derives parameters; Section 12 gives complete pseudocode; Sections 13–16 cover performance engineering, cost accounting, implementation security, and testing; Section 17 maps extensions.

2 Preliminaries

2.1 Notation

λ denotes the security parameter; the running target is $\lambda = 128$. q is an odd prime and $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$, with representatives taken in $(-q/2,\, q/2\,] \cap \mathbb{Z}$; "mod q" always means reduction to this representative set. Bold lower case ($\mathbf{x}$) denotes column vectors, upper case (A) matrices. $\|\mathbf{x}\|$ is the Euclidean norm of the representative vector in $\mathbb{Z}^d$ and $\|\mathbf{x}\|_\infty$ the max norm. For a finite set $\mathcal{S}$, $x \leftarrow \mathcal{S}$ denotes uniform sampling. $[m] = \{1, \ldots, m\}$. All logarithms are base 2. A function $f(\lambda)$ is *negligible* if $f(\lambda) = o(\lambda^{-c})$ for every constant c; we write $\mathsf{negl}(\lambda)$.

Throughout, $A \in \mathbb{Z}_q^{n\times n}$ and $B, C \in \mathbb{Z}_q^{n\times m}$; columns of B are $\mathbf{b}_1, \ldots, \mathbf{b}_m$ and rows of A are $\mathbf{a}_1^\mathsf{T}, \ldots, \mathbf{a}_n^\mathsf{T}$. Square A is assumed only for notational simplicity; nothing below requires it.

Remark 2.1 (Integer semantics). The protocols prove $C = AB$ *modulo* q. If the application intends integer matrices with $\|A\|_\infty \le 2^a$ and $\|B\|_\infty \le 2^{b_B}$, then choosing $q > 2n \cdot 2^{a+b_B}$ makes the mod-q statement equivalent to the integer statement, because entries of the true integer product cannot wrap. Note, however, that the proof does not by itself certify that the *committed* B has small entries; if that must itself be enforced against a malicious prover, lattice range proofs [21] can be layered on. In most deployments mod-q semantics are simply adopted by convention and the issue does not arise.

2.2 Interactive proofs and their three properties

Definition 2.2 (Interactive proof system)**.** An *interactive protocol* between a prover $\mathcal{P}$ (holding a statement x and a witness w) and a verifier $\mathcal{V}$ (holding x) is a sequence of messages ending with $\mathcal{V}$ outputting accept or reject. The protocol is *public-coin* if every verifier message is a fresh uniform sample from a prescribed set, independent of everything else (the verifier "flips coins in public"). All protocols in this document are public-coin.

Fix a binary relation $R = \{(x, w)\}$; the *language* is $L_R = \{x : \exists w,\ (x, w) \in R\}$. For us, x will consist of public matrices and/or commitments, and w of the hidden matrices together with commitment openings.

Definition 2.3 (Completeness)**.** The protocol is *complete* if for every $(x, w) \in R$, when both parties are honest, $\Pr[\mathcal{V} \text{ accepts}] \ge 1 - \mathsf{negl}(\lambda)$.

Definition 2.4 (Knowledge soundness)**.** The protocol is *knowledge-sound with knowledge error* κ if there is an efficient *extractor* E such that for every (possibly malicious, efficient) prover $\mathcal{P}^*$ that makes $\mathcal{V}$ accept on statement x with probability $\epsilon > \kappa$, the extractor E, given the code of $\mathcal{P}^*$ with the ability to rewind it, outputs a witness w with $(x, w) \in R$ in expected time polynomial in $\lambda/(\epsilon - \kappa)$.

Knowledge soundness implies ordinary soundness (a false statement has no witness, so acceptance probability is at most κ), and is the right notion when statements refer to committed values: it says a convincing prover doesn't merely certify a bare existential but actually *possesses* openings satisfying the relation. Note the quantifier: the guarantee is against arbitrary prover strategies, not just protocol-following ones. The rewinding in the definition — running $\mathcal{P}^*$ twice from the same internal state with different challenges — is a proof device that exists only inside the security argument; no real execution ever answers two challenges with one mask, which is what preserves zero-knowledge.

Definition 2.5 (Honest-verifier zero-knowledge (HVZK))**.** The protocol is *statistically (resp. computationally) HVZK* if there is an efficient *simulator* S that, given only the statement x, outputs a full transcript whose distribution is within negligible statistical distance of (resp. computationally indistinguishable from) the transcript of an honest execution.

HVZK is the right target here because the protocols will be deployed *non-interactively* via the Fiat–Shamir transform (Section 7.5), where the "verifier" is a hash function and is honest by construction. The conceptual content of Definition 2.5 bears emphasis for a reader new to the area: if every transcript the verifier sees could have been manufactured by the verifier alone, from public data, without ever contacting the prover, then the interaction cannot have carried information about the witness. Zero knowledge is a statement about the *distribution* of conversations, and proving it means exhibiting the simulator.

Definition 2.6 (Statistical distance)**.** For distributions P, Q on a countable set Ω, $\Delta(P, Q) = \frac{1}{2}\sum_{\omega} |P(\omega) - Q(\omega)|$. No procedure, of any computational power, can distinguish one sample of P from one of Q with advantage exceeding $\Delta(P, Q)$.

The qualifier *statistical* on any guarantee means it holds against computationally unbounded adversaries, via a statistical-distance bound; *computational* means it holds against adversaries bounded by (astronomically large but finite) computation, conditional on a hardness assumption. Keeping the two separate is essential; Section 4.2 shows that in this problem one of soundness and hiding *must* be computational.

2.3 Σ-protocols and special soundness

The engine room of both protocols is a three-move pattern.

Definition 2.7 (Σ-protocol)**.** A Σ-*protocol* for relation R is a three-move public-coin protocol: $\mathcal{P}$ sends a first message w (the *mask* or *commitment message*), $\mathcal{V}$ sends a uniform challenge $\gamma \leftarrow \mathcal{C}$, $\mathcal{P}$ sends a response z, and $\mathcal{V}$ applies a deterministic check to (x, w, γ, z).

Definition 2.8 (k-special soundness)**.** A Σ-protocol is *k-special sound* if there is an efficient algorithm that, from any k accepting transcripts $(w, \gamma_1, z_1), \ldots, (w, \gamma_k, z_k)$ sharing the same first message and having pairwise distinct challenges, computes a witness for x (possibly for a mildly relaxed relation, made precise per protocol).

k-special soundness implies knowledge soundness with knowledge error $(k-1)/|\mathcal{C}|$ by a standard rewinding argument [19]: a prover that convinces with probability noticeably above $(k-1)/|\mathcal{C}|$ can be rewound to produce k accepting transcripts on a common first message, from which the witness is computed. Intuitively, a prover with no witness can prepare a first message that answers at most $k-1$ of the $|\mathcal{C}|$ possible challenges, so it survives a random challenge with probability at most $(k-1)/|\mathcal{C}|$. Protocol Π_1 will be 2-special sound (two transcripts suffice: extraction is a divided difference); Π_2 will be 3-special sound (three transcripts: extraction interpolates a quadratic).

Our protocols are strictly speaking five-move (an initial compression challenge precedes the Σ-protocol); the framework of multi-round special soundness [19, 20] covers this, and we use its composition and Fiat–Shamir theorems as black boxes, doing the single-round arguments in full ourselves.

3 Freivalds Compression

The first ingredient converts the size-nm claim $C = AB$ into a size-n (or even scalar) claim, at the cost of a small, quantified soundness error. The idea is Freivalds' [1]: apply both sides to a random vector.

Because commitment norms will constrain us to *small-entry* projection vectors (Section 1.4), we prove the compression lemma over an arbitrary subset $\mathcal{S} \subseteq \mathbb{Z}_q$, in the style of the Schwartz–Zippel lemma [8, 9]. Uniform sampling over all of $\mathbb{Z}_q$ is the special case $\mathcal{S} = \mathbb{Z}_q$.

Lemma 3.1 (Subset Freivalds, one-sided)**.** *Let $D \in \mathbb{Z}_q^{n\times m}$ with $D \neq 0$, let $\mathcal{S} \subseteq \mathbb{Z}_q$ be finite and nonempty, and let $\mathbf{v} \leftarrow \mathcal{S}^m$ be uniform. Then* $\Pr[\, D\mathbf{v} = 0 \bmod q \,] \leq 1/|\mathcal{S}|$.

Proof. Pick a nonzero entry D_{ij^*} of D, and consider row i: the event $D\mathbf{v} = 0$ implies $\sum_j D_{ij} v_j = 0$, i.e.

$$D_{ij^*}\, v_{j^*} \;=\; -\textstyle\sum_{j\neq j^*} D_{ij} v_j \pmod q.$$

Condition on any fixed values of $(v_j)_{j\neq j^*}$; the right-hand side becomes a constant $c \in \mathbb{Z}_q$. Since q is prime and $D_{ij^*} \neq 0$, the map $v \mapsto D_{ij^*} v$ is a bijection of $\mathbb{Z}_q$, so exactly one element of $\mathbb{Z}_q$ — hence *at most one* element of $\mathcal{S}$ — satisfies the displayed equation. As v_{j^*} is uniform on $\mathcal{S}$ and independent of the conditioning, the conditional probability is at most $1/|\mathcal{S}|$, and averaging over the conditioning preserves the bound. □

Lemma 3.2 (Subset Freivalds, two-sided)**.** *Let $D \in \mathbb{Z}_q^{n\times m}$ with $D \neq 0$ and let $\mathbf{s} \leftarrow \mathcal{S}^n$, $\mathbf{v} \leftarrow \mathcal{S}^m$ be independent and uniform. Then* $\Pr[\, \mathbf{s}^{\mathsf{T}} D\mathbf{v} = 0 \bmod q \,] \leq 2/|\mathcal{S}|$.

Proof. Write the event as $\mathbf{s}^{\mathsf{T}}(D\mathbf{v}) = 0$. By Lemma 3.1, $\Pr[D\mathbf{v} = 0] \leq 1/|\mathcal{S}|$. Conditioned on $D\mathbf{v} \neq 0$, apply Lemma 3.1 again with the nonzero "matrix" $(D\mathbf{v})^{\mathsf{T}}$ and random vector $\mathbf{s}$: the conditional probability that $\mathbf{s}^{\mathsf{T}}(D\mathbf{v}) = 0$ is at most $1/|\mathcal{S}|$. A union bound finishes. □

Applied with $D = C - AB$: if the verifier ever observes (through the cryptographic machinery below) that $C\mathbf{v} = AB\mathbf{v}$ for its own fresh random $\mathbf{v}$, then either $C = AB$ or an event of probability $\leq 1/|\mathcal{S}|$ occurred. Independent repetitions multiply the error. Two remarks position this correctly.

Remark 3.3 (Why a modulus, and why a prime)*.* Working in $\mathbb{Z}_q$ rather than $\mathbb{Z}$ serves both soundness and secrecy. For soundness, the proof above needs invertibility of nonzero scalars, which primality of q supplies. For secrecy, $\mathbb{Z}_q$ is a *compact* group carrying a uniform measure: a secret masked by uniform noise is perfectly hidden (a one-time pad), whereas on the noncompact group $\mathbb{Z}$ no uniform distribution exists and every noise distribution leaks something about the magnitude of what it

masks. The informal question "how large must the masking noise be?" is dissolved, not answered, by moving to $\mathbb{Z}_q$.

Remark 3.4 (Compression is not binding). Lemma 3.1 silently assumes D is *fixed before* $\mathbf{v}$ is drawn. With hidden matrices, nothing in the linear algebra itself enforces this: a prover free to choose its matrices after seeing $\mathbf{v}$ can satisfy the projected equation exactly while $C \neq AB$ (choose any $\hat{C}$ with $\hat{C}\mathbf{v} = AB\mathbf{v}$, an affine space of dimension $n(m-1)$). The entire role of the commitments in Section 4 is to freeze D in time. Freivalds gives leverage; commitments give order-of-events.

4 Commitments, and Why Binding Must Be Computational

4.1 Definitions

Definition 4.1 (Commitment scheme)**.** A commitment scheme (in the common-reference-string model) consists of a setup algorithm producing public parameters crs, and a commitment function $\mathsf{com}_{\mathsf{crs}}(\mathbf{x}; \boldsymbol{\rho})$ mapping a message $\mathbf{x}$ and randomness $\boldsymbol{\rho}$ (drawn from a prescribed distribution) to a commitment string t. An *opening* of t is a pair $(\mathbf{x}, \boldsymbol{\rho})$, subject to a validity predicate, with $\mathsf{com}_{\mathsf{crs}}(\mathbf{x}; \boldsymbol{\rho}) = t$. The scheme is:

- *hiding* if commitments to any two messages are indistinguishable (statistically or computationally) given crs;
- *binding* if no adversary can produce one commitment t with two valid openings $(\mathbf{x}, \boldsymbol{\rho})$, $(\mathbf{x}', \boldsymbol{\rho}')$, $\mathbf{x} \neq \mathbf{x}'$ (statistically: no such pair exists except with negligible probability over crs; computationally: finding one is infeasible);
- *additively homomorphic* if $\mathsf{com}(\mathbf{x}; \boldsymbol{\rho}) + \mathsf{com}(\mathbf{x}'; \boldsymbol{\rho}') = \mathsf{com}(\mathbf{x} + \mathbf{x}';\ \boldsymbol{\rho} + \boldsymbol{\rho}')$ and $c \cdot \mathsf{com}(\mathbf{x}; \boldsymbol{\rho}) = \mathsf{com}(c\mathbf{x};\ c\boldsymbol{\rho})$ for scalars c, where addition of commitment strings is coordinatewise in $\mathbb{Z}_q$.

Homomorphism is what lets the verifier operate on sealed containers: given commitments to the columns $\mathbf{b}_1, \ldots, \mathbf{b}_m$ of B, anyone can compute a commitment to $B\mathbf{v} = \sum_j v_j \mathbf{b}_j$ without opening anything. The random projection of Section 3 thus lifts from matrices to commitments.

4.2 Why binding is necessary at all

Two propositions locate exactly where cryptographic hardness enters this problem. The first says that with no binding there is no statement; the second says that binding and hiding cannot both be unconditional, so a hardness assumption is forced.

Proposition 4.2 (Vacuity without binding)**.** *Fix public $A \in \mathbb{Z}_q^{n \times n}$ invertible, and consider any interactive protocol in which the verifier's view is some function of quantities of the form $\{$masked linear images of $B\}$ with no binding artifact fixed in advance. Then for* every *pair $(\hat{B}, \hat{C})$ with $\hat{C} = A\hat{B}$, and every transcript T the honest prover could produce for its true (B, C), there is a prover strategy holding $(\hat{B}, \hat{C})$ producing T with the same probability. Consequently the protocol's accepting transcripts certify only membership in $\{(B', C') : C' = AB'\}$ — a set containing an element for every possible C' (namely $B' = A^{-1}C'$) — and the "statement" carries no information restricting the hidden matrices.*

Proof. The relation $\{(B', C') : C' = AB'\}$ projects surjectively onto each coordinate: for any C', the pair $(A^{-1}C', C')$ lies in it. A prover holding $(\hat{B}, \hat{C})$ on the relation simply runs the honest protocol for its own pair. Without a binding artifact created before the interaction, no function of

the transcript distinguishes which relation member was used beyond what the protocol deliberately reveals, which by zero-knowledge is nothing beyond membership. Membership is universally satisfiable, so it constrains nothing. □

The content of Proposition 4.2 is not that such a protocol is *insecure*; it is that it is *about nothing*. "There exist matrices multiplying correctly" is a tautology. The meaningful statement is "*the* matrices sealed in *these* commitments multiply correctly," and that sentence cannot be uttered without commitments. This is the formal answer to the natural question of whether clever masking alone — large noise, random scalings — could replace cryptographic binding: masking addresses secrecy, but the soundness gap is not a secrecy problem, it is an *identity-over-time* problem.

Proposition 4.3 (No commitment is both statistically hiding and statistically binding)**.** *Suppose a commitment scheme over message space $\mathcal{M}$, $|\mathcal{M}| \geq 2$, is statistically binding. Then it is not statistically hiding: an unbounded adversary distinguishes commitments to distinct messages with advantage $1 - \mathsf{negl}(\lambda)$.*

Proof. Statistical binding means that, except with negligible probability over crs, the map from a commitment string to the message it can be validly opened to is well defined (single valued) on the support of honestly generated commitments: for each attainable t, valid openings of t agree on the message. Fix such a crs, and let $\mathsf{Supp}(\mathbf{x})$ denote the set of strings attainable as $\mathsf{com}(\mathbf{x}; \cdot)$. Single valuedness makes $\{\mathsf{Supp}(\mathbf{x})\}_{\mathbf{x}}$ pairwise disjoint. An unbounded distinguisher, given t, enumerates all $(\mathbf{x}, \boldsymbol{\rho})$ and outputs the unique $\mathbf{x}$ whose support contains t; it errs only on the negligible-probability bad crs event. Hence commitments to distinct messages are almost perfectly distinguishable — the negation of statistical hiding. □

Corollary 4.4 (One unconditional guarantee per side)**.** *In any commitment-based protocol for hidden matrix multiplication, at most one of {secrecy of the matrices, impossibility of equivocation} can hold against unbounded adversaries. The other must rest on a computational assumption. The construction below offers both assignments: statistically hiding with computationally binding (secrecy unconditional, forgery 2^{128}-hard), or computationally hiding with computationally binding at much lower cost (both guarantees 2^{128}-hard). Section 5.4 quantifies the price of the first option.*

There is also nothing to fall back on: revealing B to get statistical soundness "the information-theoretic way" is exactly what the problem statement forbids. A hardness assumption is therefore not an implementation shortcut in this problem; it is load-bearing, and the only freedom is where to place it.

5 The Lattice Commitment Scheme

5.1 Hardness assumptions

Definition 5.1 (SIS: Short Integer Solution)**.** For parameters (k, N, q, β), the problem $\mathsf{SIS}_{k,N,q,\beta}$ is: given a uniform matrix $G \leftarrow \mathbb{Z}_q^{k \times N}$, find $\mathbf{u} \in \mathbb{Z}^N$ with $G\mathbf{u} = 0 \bmod q$, $\mathbf{u} \neq 0$, and $\|\mathbf{u}\| \leq \beta$.

Solutions always exist once $\beta \gtrsim \sqrt{N}\, q^{k/N}$ (pigeonhole / Minkowski); the assumption is that *finding* one is infeasible for suitable parameters. Ajtai's theorem [6] gives the structural reason for confidence: solving SIS for *random* G, even on average, is at least as hard as solving worst-case approximate short-vector problems on *every* lattice of dimension N. An average-case break would collapse a whole worst-case geometric problem class studied since Minkowski. Concrete cost estimates for the best known algorithms are in Section 9.

Definition 5.2 (Knapsack LWE)**.** For $M < N$ and a short-entry distribution χ (here: uniform ternary $\{-1, 0, 1\}$), the problem $\mathsf{kLWE}_{M,N,q,\chi}$ is: distinguish $(G,\, G\boldsymbol{\rho})$ from $(G,\, \mathbf{u})$, where $G \leftarrow \mathbb{Z}_q^{M\times N}$, $\boldsymbol{\rho} \leftarrow \chi^N$, $\mathbf{u} \leftarrow \mathbb{Z}_q^M$.

The condition $M < N$ makes $\boldsymbol{\rho} \mapsto G\boldsymbol{\rho}$ compressing, so $G\boldsymbol{\rho}$ has exponentially many short preimages; the assumption is that it looks uniform. This "knapsack" form is polynomial-time equivalent to the standard decisional Learning-With-Errors problem in dimension $N - M$ [12, 7]; the surplus dimension $N - (k + n)$ will be the security dial for computational hiding. (Note the sanity check hidden in the inequality: if $M \geq N$ the system $G\boldsymbol{\rho} = \mathbf{y}$ is generically uniquely solvable by linear algebra and pseudorandomness is simply false. The commitment dimensions below always keep the randomness strictly longer than the output.)

5.2 The construction

Construction 5.3 (Ajtai/BDLOP-style commitment [6, 15])**.** Public parameters (crs, expanded from a public seed; Section 12): uniform $G_1 \in \mathbb{Z}_q^{k\times N}$ and $G_2 \in \mathbb{Z}_q^{n\times N}$. Message space $\mathbb{Z}_q^n$; randomness $\boldsymbol{\rho} \leftarrow \{-1, 0, 1\}^N$. Define

$$\mathsf{com}(\mathbf{x}; \boldsymbol{\rho}) \;=\; \big(G_1\boldsymbol{\rho},\;\; G_2\boldsymbol{\rho} + \mathbf{x}\big) \;\in\; \mathbb{Z}_q^k \times \mathbb{Z}_q^n,$$

written $t = (t_1, t_2)$. A *valid opening* of t is a pair $(\mathbf{x}, \boldsymbol{\rho})$ with the equation holding and $\|\boldsymbol{\rho}\| \leq B_{\text{open}}$ for a bound fixed by the protocol. The scheme is additively homomorphic in the sense of Definition 4.1, by linearity of both slots.

The two slots have distinct jobs. The first slot $G_1\boldsymbol{\rho}$ never touches the message: it is a *fingerprint of the randomness*, and (as the binding lemma below shows) it pins $\boldsymbol{\rho}$ down up to SIS-hard ambiguity, after which the second slot determines $\mathbf{x} = t_2 - G_2\boldsymbol{\rho}$. Crucially, only the *randomness* is required to be short; the message ranges over all of $\mathbb{Z}_q^n$. This asymmetry drives the protocol design: message-valued secrets can be masked with uniform noise (perfect hiding), while randomness-valued secrets must be masked with care to preserve shortness (Section 6).

5.3 Binding, including the relaxed form the protocols actually need

Protocol extraction (Sections 7–8) never produces exact openings; it produces openings scaled by a small *slack* factor (a difference or product of differences of challenges). The right formalism is standard [15, 18]:

Definition 5.4 (Relaxed opening)**.** A *relaxed opening* of $t = (t_1, t_2)$ with slack bound $\bar{T}$ and norm bound B is a triple $(\mathbf{x}, \boldsymbol{\rho}, \bar{c})$ with $\bar{c} \in \mathbb{Z}$, $0 < |\bar{c}| \leq \bar{T}$, $\|\boldsymbol{\rho}\| \leq B$, and

$$G_1\boldsymbol{\rho} = \bar{c}\, t_1 \pmod q, \qquad G_2\boldsymbol{\rho} + \bar{c}\,\mathbf{x} = \bar{c}\, t_2 \pmod q.$$

An exact opening is the case $\bar{c} = 1$.

Lemma 5.5 (Relaxed binding from SIS)**.** *Suppose $(\mathbf{x}, \boldsymbol{\rho}, \bar{c})$ and $(\mathbf{x}', \boldsymbol{\rho}', \bar{c}')$ are relaxed openings of the same t, with slack bounds $\bar{T}, \bar{T}'$ and norm bounds B, B' respectively, and $\mathbf{x} \neq \mathbf{x}' \bmod q$. Then $\mathbf{u} := \bar{c}'\boldsymbol{\rho} - \bar{c}\,\boldsymbol{\rho}'$ is a solution to $\mathsf{SIS}_{k,N,q,\beta}$ for G_1, with $\beta = \bar{T}'B + \bar{T}B'$.*

Proof. First-slot equations give $G_1\mathbf{u} = \bar{c}'(\bar{c}\, t_1) - \bar{c}(\bar{c}'\, t_1) = 0 \bmod q$. Second-slot equations give

$$G_2\mathbf{u} \;=\; \bar{c}'\big(\bar{c}t_2 - \bar{c}\mathbf{x}\big) - \bar{c}\big(\bar{c}'t_2 - \bar{c}'\mathbf{x}'\big) \;=\; \bar{c}\,\bar{c}'\,(\mathbf{x}' - \mathbf{x}) \pmod q.$$

Since q is prime, $0 < |\bar{c}|, |\bar{c}'| < q$ makes $\bar{c}\bar{c}' \neq 0 \bmod q$, and $\mathbf{x}' \neq \mathbf{x}$ then forces $G_2\mathbf{u} \neq 0 \bmod q$, hence $\mathbf{u} \neq 0 \bmod q$, hence $\mathbf{u} \neq 0$ as an integer vector. The norm bound is the triangle inequality: $\|\mathbf{u}\| \leq |\bar{c}'|\|\boldsymbol{\rho}\| + |\bar{c}|\|\boldsymbol{\rho}'\| \leq \bar{T}'B + \bar{T}B'$. □

In words: the only way to make one container open two ways — even in the relaxed, slack-scaled sense — is to know a short nonzero integer kernel vector of the random matrix G_1. The protocol parameterization (Part II) will compute the largest β any extraction argument needs and size (k, N, q) so that $\mathsf{SIS}_{k,N,q,\beta}$ costs $\geq 2^{128}$ operations.

5.4 Hiding, in two modes

Hiding of com reduces to the pseudorandomness (or statistical uniformity) of the pair $(G_1\boldsymbol{\rho}, G_2\boldsymbol{\rho})$, since adding an independent uniform-looking vector to any fixed message hides it. Write $G \in \mathbb{Z}_q^{(k+n)\times N}$ for the stack of G_1, G_2.

Lemma 5.6 (Statistical hiding via the leftover hash lemma)**.** *If*

$$N \log_2 3 \;\geq\; (k+n)\log_2 q \;+\; 2\lambda + 2,$$

then for G uniform and $\boldsymbol{\rho} \leftarrow \{-1, 0, 1\}^N$, the distribution of $(G,\, G\boldsymbol{\rho})$ is within statistical distance $2^{-\lambda}$ of $(G,\, \mathsf{uniform})$. Consequently the commitment is statistically hiding: no adversary of any computational power learns anything about $\mathbf{x}$ from $\mathsf{com}(\mathbf{x}; \boldsymbol{\rho})$ beyond $2^{-\lambda}$ advantage.

Proof sketch. The family $h_G(\boldsymbol{\rho}) = G\boldsymbol{\rho}$ is universal: for $\boldsymbol{\rho} \neq \boldsymbol{\rho}'$, $\Pr_G[G\boldsymbol{\rho} = G\boldsymbol{\rho}'] = q^{-(k+n)}$, since $G(\boldsymbol{\rho} - \boldsymbol{\rho}')$ is uniform for any fixed nonzero difference. The leftover hash lemma [4] then bounds the distance of $(G, h_G(\boldsymbol{\rho}))$ from uniform by $\frac{1}{2}\sqrt{2^{-H_\infty(\boldsymbol{\rho})}\, q^{k+n}}$, and $H_\infty(\boldsymbol{\rho}) = N \log_2 3$ for uniform ternary $\boldsymbol{\rho}$. The stated inequality makes the bound $\leq 2^{-\lambda}$. □

Lemma 5.7 (Computational hiding via knapsack LWE)**.** *If $k + n < N$, then any distinguisher between commitments to chosen messages $\mathbf{x}_0, \mathbf{x}_1$ with advantage δ yields a distinguisher for $\mathsf{kLWE}_{k+n,\, N,\, q,\, \mathit{ternary}}$ with advantage $\delta/2$.*

Proof. Under kLWE, replace $(G, G\boldsymbol{\rho})$ by $(G, \mathbf{u})$ with $\mathbf{u}$ uniform; then $\mathsf{com}(\mathbf{x}_b; \boldsymbol{\rho}) = (\mathbf{u}_1, \mathbf{u}_2 + \mathbf{x}_b)$ is exactly uniform, independent of b. Standard hybrid argument. □

The two modes are interchangeable at the protocol level — every theorem below invokes hiding as a black box — but wildly different in cost. Statistical hiding needs the randomness entropy $N \log_2 3$ to *exceed the total output length in bits*, which for word-sized q and dimensions in the low thousands puts N near 10^5; computational hiding needs only $N - (k+n)$ to be an LWE-secure dimension, an order of magnitude smaller. Part II carries both through the numbers; the reference instantiation uses the computational mode, and the statistical surcharge is reported for the deployment that wants unconditional secrecy of B (compare Corollary 4.4: in the computational mode, *both* secrecy and binding are 2^{128}-hardness claims about lattices; in the statistical mode, secrecy holds against unbounded adversaries and only forgery is computational).

Remark 5.8 (Which mode matches which threat model). Choose the statistical mode when the hidden matrices must stay secret *forever*, against adversaries with unbounded future compute (harvest-now-decrypt-later concerns for data whose sensitivity outlives all technology forecasts). Choose the computational mode when "2^{128} lattice operations, quantum included, per Section 9" is an acceptable secrecy bar — which matches essentially all commercial threat models, and is the mode production lattice systems [15, 21, 25] use.

5.5 Well-formedness of the statement commitments

The soundness theorems below compare an opening extracted from the protocol against "the" committed matrices, which presumes each statement commitment $T^{(j)}$ *has* a well-defined committed value — i.e., that the prover knows some valid (possibly relaxed) opening of it. Three ways to discharge this, in decreasing order of cryptographic strength:

1. *Proofs of opening at commitment time.* A one-time Σ-protocol per commitment proving knowledge of short $\boldsymbol{\rho}$ with $G_1\boldsymbol{\rho} = t_1$ (the message slot is then determined). This is a strictly simpler special case of protocol Π_1's Σ-step and is specified in Section 12. Amortized versions proving all m columns at sublinear cost exist [15, 18, 22]; their bookkeeping is out of scope here.

2. *Later-audit deployments.* If the committer will eventually open the commitments to an auditor (e.g. a model owner whose weights are escrowed), malformed commitments are self-defeating and the opening proofs can be skipped; binding alone carries the protocol.

3. *Same-party deployments.* When the party that created the commitments is the same party that later verifies proofs against them (or a party that received the openings out of band), well-formedness is trivially known.

Throughout Part I we state theorems for statements whose commitments are well formed in the above sense, with slack bound $\bar{T}_0$ and randomness bound B_0 for the known openings ($\bar{T}_0 = 1$, $B_0 = \sqrt{N}$ for honestly generated ternary randomness); the extraction arguments track these symbols so the composition with opening proofs (which yield $\bar{T}_0 > 1$) is mechanical.

6 Masking: Uniform Noise and Rejection Sampling

The prover's responses will have the shape mask $+ \gamma \cdot$ secret. Two regimes arise, matching the two slots of the commitment.

6.1 Message slot: uniform masking is perfect

Lemma 6.1 (One-time pad over $\mathbb{Z}_q$)**.** *Fix any $\mathbf{b} \in \mathbb{Z}_q^n$ and any $\gamma \in \mathbb{Z}_q$. If $\boldsymbol{\varepsilon} \leftarrow \mathbb{Z}_q^n$ is uniform, then $\mathbf{z} = \boldsymbol{\varepsilon} + \gamma\mathbf{b} \bmod q$ is uniform on $\mathbb{Z}_q^n$ and independent of $(\mathbf{b}, \gamma)$.*

Proof. For each fixed $(\mathbf{b}, \gamma)$ the map $\boldsymbol{\varepsilon} \mapsto \boldsymbol{\varepsilon} + \gamma\mathbf{b}$ is a bijection of the finite group $\mathbb{Z}_q^n$, and bijections preserve the uniform distribution. □

This is why message-valued responses cost nothing to protect (compare Remark 3.3). Each mask $\boldsymbol{\varepsilon}$ is used for exactly one response; a reused mask allows differencing and is fatal (Section 15).

6.2 Randomness slot: Gaussian masking with rejection

Randomness-valued responses $\boldsymbol{\zeta} = \boldsymbol{\eta} + \gamma\boldsymbol{\rho}$ must be *short* (their shortness is what the binding argument consumes), so uniform-mod-q masking is unavailable. Bounded noise leaks at the boundary: if $\boldsymbol{\eta}$ is uniform on a box and the observed $\boldsymbol{\zeta}$ lies at an extreme, the possible values of $\gamma\boldsymbol{\rho}$ are constrained. The standard resolution [10, 11] makes the *output* distribution independent of the secret by construction:

Definition 6.2 (Discrete Gaussian)**.** $\mathcal{D}_\sigma^N$ is the distribution on $\mathbb{Z}^N$ with $\Pr[\mathbf{y}] \propto \exp(-\|\mathbf{y}\|^2/2\sigma^2)$.

Lemma 6.3 (Rejection sampling [11]). *Let $\mathbf{c} \in \mathbb{Z}^N$ with $\|\mathbf{c}\| \le T$ (the* center*: in our use, $\gamma\boldsymbol{\rho}$-type terms), let $\sigma = \alpha T$, and let $M = \exp\left(\frac{12}{\alpha} + \frac{1}{2\alpha^2}\right)$. Consider the procedure: sample $\mathbf{y} \leftarrow \mathcal{D}_\sigma^N$, set $\boldsymbol{\zeta} = \mathbf{y} + \mathbf{c}$, output $\boldsymbol{\zeta}$ with probability*

$$p_{\mathrm{acc}}(\boldsymbol{\zeta}) = \min\left(1,\ \frac{\mathcal{D}_\sigma^N(\boldsymbol{\zeta})}{M \cdot \mathcal{D}_\sigma^N(\boldsymbol{\zeta} - \mathbf{c})}\right) = \min\left(1,\ \frac{1}{M}\exp\left(\frac{\|\mathbf{c}\|^2 - 2\langle \boldsymbol{\zeta},\, \mathbf{c}\rangle}{2\sigma^2}\right)\right),$$

otherwise abort. Then the procedure outputs something with probability $\ge (1 - 2^{-100})/M$, and conditioned on producing output, the distribution of $\boldsymbol{\zeta}$ is within statistical distance 2^{-100} of $\mathcal{D}_\sigma^N$ — in particular, independent of $\mathbf{c}$.

We fix $\alpha = 12$ throughout, giving $M = e^{1+1/288} \approx 2.76$ expected attempts. Geometrically: the shifted Gaussian is pointwise dominated by M times the centered one except far in the tail, so thinning the shifted samples reproduces the centered law exactly; the secret shift is erased, not merely drowned.

Lemma 6.4 (Tail bound [5]). *For $\boldsymbol{\zeta} \leftarrow \mathcal{D}_\sigma^N$ and any $c \ge 1$: $\Pr[\|\boldsymbol{\zeta}\| > c\,\sigma\sqrt{N}\,] \le c^N e^{N(1-c^2)/2}$. With $c = 1.2$ this is $e^{-0.0377N}$, negligible for all dimensions used here.*

The verifier will therefore enforce $\|\boldsymbol{\zeta}\| \le B := 1.2\,\sigma\sqrt{N}$ (for the appropriate dimension); honest responses pass except with negligible probability, and the bound is what enters the SIS parameter via Lemma 5.5. When several randomness responses are produced in one round, they are treated as a single concatenated vector and rejected jointly, so the expected number of restarts stays M rather than $M^{\#\text{responses}}$.

Remark 6.5 (Flooding, for contrast). The older alternative — noise wider than the secret by a 2^λ factor, no rejection — gives the same secrecy with per-signature simplicity but adds λ bits to every norm, hence to q and to all dimensions. Rejection sampling is the reason modern lattice signatures are compact, and we adopt it; the flooding variant remains a valid fallback anywhere Lemma 6.3 is cited.

7 Protocol Π_1: Public A, Hidden B and C

7.1 Statement and parameters

Public: $A \in \mathbb{Z}_q^{n\times n}$; commitments $T_B^{(j)} = \mathsf{com}(\mathbf{b}_j; \boldsymbol{\rho}_j)$ and $T_C^{(j)} = \mathsf{com}(\mathbf{c}_j; \boldsymbol{\tau}_j)$ for $j \in [m]$, well formed in the sense of Section 5.5 with $(\bar{T}_0, B_0)$. Claimed: the committed matrices satisfy $C = AB \bmod q$. Parameters: coefficient set $\mathcal{S}_b = \{0, 1, \ldots, 2^b - 1\} \subset \mathbb{Z}_q$ used both for compression vectors and challenges ($\mathcal{C} = \mathcal{S}_b$); Gaussian width σ_1 and response bound $B_1 = 1.2\,\sigma_1\sqrt{2N}$ from Section 6; the protocol below is one repetition, run r times in parallel (Section 7.5).

Protocol Π_1 (one repetition).

V1. $\mathcal{V}$ sends a compression vector $\mathbf{v} \leftarrow \mathcal{S}_b^m$.
Both parties compute the homomorphic combinations

$$T_b = \sum_{j=1}^{m} v_j\, T_B^{(j)}, \qquad T_c = \sum_{j=1}^{m} v_j\, T_C^{(j)}.$$

$\mathcal{P}$ computes $\mathbf{b} = B\mathbf{v}$, $\boldsymbol{\rho}_v = \sum_j v_j\boldsymbol{\rho}_j$, $\boldsymbol{\tau}_v = \sum_j v_j\boldsymbol{\tau}_j$ (over $\mathbb{Z}$; note $\|\boldsymbol{\rho}_v\|, \|\boldsymbol{\tau}_v\| \le B_\rho :=$

$m\,2^b B_0$).

P1. $\mathcal{P}$ samples $\boldsymbol{\varepsilon} \leftarrow \mathbb{Z}_q^n$ uniform and $(\boldsymbol{\eta}, \boldsymbol{\mu}) \leftarrow \mathcal{D}_{\sigma_1}^{2N}$, and sends

$$w = \mathsf{com}(\boldsymbol{\varepsilon};\, \boldsymbol{\eta}), \qquad u = \mathsf{com}(A\boldsymbol{\varepsilon} \bmod q;\, \boldsymbol{\mu}).$$

V2. $\mathcal{V}$ sends a challenge $\gamma \leftarrow \mathcal{C} = \mathcal{S}_b$.

P2. $\mathcal{P}$ computes

$$\mathbf{z} = \boldsymbol{\varepsilon} + \gamma\,\mathbf{b} \bmod q, \qquad \boldsymbol{\zeta} = \boldsymbol{\eta} + \gamma\,\boldsymbol{\rho}_v, \qquad \boldsymbol{\xi} = \boldsymbol{\mu} + \gamma\,\boldsymbol{\tau}_v \quad (\text{over } \mathbb{Z}),$$

applies rejection sampling (Lemma 6.3) to the concatenation $(\boldsymbol{\zeta}, \boldsymbol{\xi})$ with center $(\gamma\boldsymbol{\rho}_v, \gamma\boldsymbol{\tau}_v)$; on abort, restarts from **P1** with entirely fresh $(\boldsymbol{\varepsilon}, \boldsymbol{\eta}, \boldsymbol{\mu})$. On accept, sends $(\mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$.

V3. $\mathcal{V}$ accepts iff $\|(\boldsymbol{\zeta}, \boldsymbol{\xi})\| \le B_1$ and

$$\begin{aligned} \big(G_1\boldsymbol{\zeta},\ \ G_2\boldsymbol{\zeta} + \mathbf{z}\big) &= w + \gamma\,T_b \pmod q, && (\Pi_1\text{-i}) \\ \big(G_1\boldsymbol{\xi},\ \ G_2\boldsymbol{\xi} + A\mathbf{z}\big) &= u + \gamma\,T_c \pmod q. && (\Pi_1\text{-ii}) \end{aligned}$$

The design in one breath: $\mathbf{v}$ compresses the matrix claim to the vector claim $A(B\mathbf{v}) = C\mathbf{v}$; the pair (w, u) commits to a masked probe $\boldsymbol{\varepsilon}$ *and* to its image $A\boldsymbol{\varepsilon}$; the challenge γ forces the prover to reveal the probe shifted by γ times the secret; equation (Π_1-i) certifies that $\mathbf{z}$ really is $\boldsymbol{\varepsilon} + \gamma B\mathbf{v}$ for the committed B, and equation (Π_1-ii) certifies that $A\mathbf{z}$ matches $A\boldsymbol{\varepsilon} + \gamma\,C\mathbf{v}$ for the committed C — which can only happen for both challenges of a rewound prover if $AB\mathbf{v} = C\mathbf{v}$.

7.2 Completeness

Theorem 7.1 (Completeness of Π_1)**.** *If $C = AB \bmod q$ and both parties are honest, $\mathcal{V}$ accepts with probability $1 - \mathsf{negl}(\lambda)$; the prover restarts* P1 *an expected $M \approx 2.76$ times.*

Proof. First slots: $G_1\boldsymbol{\zeta} = G_1\boldsymbol{\eta} + \gamma G_1\boldsymbol{\rho}_v$, and $w_1 + \gamma T_{b,1} = G_1\boldsymbol{\eta} + \gamma G_1\boldsymbol{\rho}_v$ by homomorphy; likewise for $\boldsymbol{\xi}$. Second slot of (Π_1-i): $G_2\boldsymbol{\zeta} + \mathbf{z} = (G_2\boldsymbol{\eta} + \boldsymbol{\varepsilon}) + \gamma(G_2\boldsymbol{\rho}_v + B\mathbf{v}) = w_2 + \gamma T_{b,2}$. Second slot of ($\Pi_1$-ii):

$$G_2\boldsymbol{\xi} + A\mathbf{z} = (G_2\boldsymbol{\mu} + A\boldsymbol{\varepsilon}) + \gamma\big(G_2\boldsymbol{\tau}_v + AB\mathbf{v}\big), \qquad u_2 + \gamma T_{c,2} = (G_2\boldsymbol{\mu} + A\boldsymbol{\varepsilon}) + \gamma\big(G_2\boldsymbol{\tau}_v + C\mathbf{v}\big),$$

equal exactly when $AB\mathbf{v} = C\mathbf{v} \bmod q$, which holds since $C = AB$. The norm check passes by Lemma 6.4, since accepted $(\boldsymbol{\zeta}, \boldsymbol{\xi})$ is 2^{-100}-close to $\mathcal{D}_{\sigma_1}^{2N}$; the restart count is Lemma 6.3. The center norm is $\|(\gamma\boldsymbol{\rho}_v, \gamma\boldsymbol{\tau}_v)\| \le \sqrt{2}\,2^b B_\rho =: T_1$, so $\sigma_1 = 12\,T_1$ satisfies the lemma's hypothesis. □

7.3 Knowledge soundness

Theorem 7.2 (Special soundness of Π_1)**.** *Fix $\mathbf{v}$ and two accepting transcripts with common first message (w, u) and distinct challenges $\gamma \neq \gamma'$. Then either the extractor computes, in one pass of linear algebra, an explicit solution to $\mathsf{SIS}_{k,N,q,\beta_1}$ for G_1 with*

$$\beta_1 \;=\; \bar{T}_0 \cdot 2B_1 \;+\; 2^b \cdot B_\rho,$$

or the committed matrices satisfy $(C - AB)\,\mathbf{v} = 0 \bmod q$. Consequently one repetition of Π_1 has knowledge error

$$\kappa_1 \;\le\; \frac{1}{2^b} \;+\; \frac{1}{2^b} \;+\; \mathsf{Adv}^{\mathsf{SIS}} \;=\; 2^{-(b-1)} + \mathsf{Adv}^{\mathsf{SIS}},$$

the two 2^{-b} terms being the Freivalds miss (Lemma 3.1 applied to $D = C - AB$) and the 2-special-soundness challenge-guessing term.

Proof. Write $\bar{c} = \gamma - \gamma'$, so $0 < |\bar{c}| < 2^b$. Subtracting the two instances of (Π_1-i):

$$\big(G_1(\boldsymbol{\zeta} - \boldsymbol{\zeta}'),\ \ G_2(\boldsymbol{\zeta} - \boldsymbol{\zeta}') + (\mathbf{z} - \mathbf{z}')\big) = \bar{c}\, T_b \pmod q.$$

Setting $\mathbf{b}^* := \bar{c}^{-1}(\mathbf{z} - \mathbf{z}') \bmod q$ (inverse in $\mathbb{Z}_q$; only the *message* is divided, never the short vector), the triple $(\mathbf{b}^*,\ \boldsymbol{\zeta} - \boldsymbol{\zeta}',\ \bar{c})$ is a relaxed opening of T_b with slack $\leq 2^b$ and norm $\leq 2B_1$. Separately, well-formedness gives relaxed openings $(\mathbf{b}_j, \boldsymbol{\rho}_j, \bar{c}_0)$ of each $T_B^{(j)}$ with $|\bar{c}_0| \leq \bar{T}_0$, $\|\boldsymbol{\rho}_j\| \leq B_0$; by homomorphy, $(B\mathbf{v},\ \boldsymbol{\rho}_v,\ \bar{c}_0)$ is a relaxed opening of T_b with slack $\leq \bar{T}_0$ and norm $\leq B_\rho$. If $\mathbf{b}^* \neq B\mathbf{v} \bmod q$, Lemma 5.5 outputs a SIS solution of norm $\leq \bar{T}_0 \cdot 2B_1 + 2^b B_\rho = \beta_1$. Otherwise $\mathbf{b}^* = B\mathbf{v}$.

The same subtraction on (Π_1-ii) gives $\big(G_1(\boldsymbol{\xi} - \boldsymbol{\xi}'),\, G_2(\boldsymbol{\xi} - \boldsymbol{\xi}') + A(\mathbf{z} - \mathbf{z}')\big) = \bar{c}\, T_c$, i.e. a relaxed opening of T_c to the message $\bar{c}^{-1}A(\mathbf{z} - \mathbf{z}') = A\mathbf{b}^* = AB\mathbf{v} \bmod q$, with the same bounds. Comparing against the well-formed opening $(C\mathbf{v}, \boldsymbol{\tau}_v, \bar{c}_0)$ of T_c via Lemma 5.5: either a SIS_{β_1} solution appears, or $C\mathbf{v} = AB\mathbf{v} \bmod q$.

For the knowledge error: condition on no SIS break. If $(C - AB)\mathbf{v} \neq 0$ — an event of probability $\geq 1 - 2^{-b}$ over $\mathbf{v}$ when $C \neq AB$, by Lemma 3.1 — then by the above no first message can answer two distinct challenges, so the prover convinces with probability at most $1/|\mathcal{C}| = 2^{-b}$. The union of the two failure sources plus $\mathsf{Adv}^{\mathsf{SIS}}$ bounds κ_1; the extractor meeting Definition 2.4 is the standard rewinding extractor for 2-special-sound protocols [19]. □

Remark 7.3 (What the slack costs). Extraction hands us $C\mathbf{v} = AB\mathbf{v}$ *on the nose* in $\mathbb{Z}_q$ — slack factors cancel in the message slot because $\mathbb{Z}_q$ is a field. Slack survives only in the SIS *norm* β_1, i.e. in parameter sizes, not in the extracted statement. This is the pleasant feature of prime-field messages that partially compensates for the parallel repetitions the small challenge sets force on us.

7.4 Zero knowledge

Theorem 7.4 (HVZK of Π_1)**.** *There is a simulator that, given only the statement $(A, \{T_B^{(j)}, T_C^{(j)}\})$ and the public coins $(\mathbf{v}, \gamma)$, produces transcripts within statistical distance 2^{-99} of honest ones. The simulation is statistical* regardless of the commitment hiding mode.

Proof. Simulator: sample $\gamma \leftarrow \mathcal{C}$; $\mathbf{z} \leftarrow \mathbb{Z}_q^n$ uniform; $(\boldsymbol{\zeta}, \boldsymbol{\xi}) \leftarrow \mathcal{D}_{\sigma_1}^{2N}$ (retrying until the norm check passes — i.e. the honest accepted law); then *define*

$$w := \big(G_1\boldsymbol{\zeta},\ G_2\boldsymbol{\zeta} + \mathbf{z}\big) - \gamma T_b, \qquad u := \big(G_1\boldsymbol{\xi},\ G_2\boldsymbol{\xi} + A\mathbf{z}\big) - \gamma T_c \pmod q,$$

and output $(\mathbf{v}, w, u, \gamma, \mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$. Compare distributions of the tuple $(\gamma, \mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$: γ is uniform in both; in the real protocol $\mathbf{z}$ is uniform and independent of everything else by Lemma 6.1; the real accepted $(\boldsymbol{\zeta}, \boldsymbol{\xi})$ is within 2^{-100} of $\mathcal{D}_{\sigma_1}^{2N}$ and independent of the secrets by Lemma 6.3. Finally, in *both* real and simulated transcripts, (w, u) is the deterministic function of $(\gamma, \mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$ displayed above — in the real protocol because the verification equations hold with equality. Identical conditionals on statistically-close marginals give statistically-close joints. □

Remark 7.5 (Where hiding is actually used). Theorem 7.4 did not invoke commitment hiding at all: the transcript leaks nothing *given the statement*. Hiding is consumed one level up, by the statement itself — the long-lived commitments $T_B^{(j)}, T_C^{(j)}$ must not leak B, C — via Lemma 5.6 or 5.7. Keeping the two layers separate is what lets the same protocol serve both hiding modes.

7.5 Repetition and the Fiat–Shamir transform

One repetition has knowledge error $\approx 2^{-(b-1)}$, far from 2^{-128} for word-friendly b. Two composition steps close the gap.

Parallel repetition. Run r independent copies (fresh $\mathbf{v}^{(\ell)}, \boldsymbol{\varepsilon}^{(\ell)}, \boldsymbol{\eta}^{(\ell)}, \boldsymbol{\mu}^{(\ell)}, \gamma^{(\ell)}$; the Step-0 commitments are shared). For multi-round special-sound protocols, r-fold parallel repetition raises the knowledge error to essentially κ_1^r [19]. The masks are what make this safe for zero knowledge: each repetition's simulator runs independently.

Fiat–Shamir. Replace $\mathcal{V}$'s coins by hash outputs: $\mathbf{v}^{(\ell)} = H(\text{crs}, \text{statement}, \ell, \texttt{"v"})$ and $\gamma^{(\ell)} = H(\text{crs}, \text{statement}, \text{all first messages}, \ell, \texttt{"gamma"})$, with H modeled as a random oracle. The proof becomes a single string, verifiable offline. A dishonest prover may now *grind*: re-randomize its first message to reroll challenges, once per hash query. For a 2μ+1-move special-sound protocol the generic bound is that Q queries inflate the cheating probability by at most a factor $\approx (Q+1)^\mu$ [20]; our protocols have $\mu = 2$ challenge rounds ($\mathbf{v}$ then γ). We therefore size

$$(Q+1)^2 \cdot \kappa_1^r \;\le\; 2^{-\lambda}, \qquad Q = 2^{64} \text{ hash-query budget},$$

i.e. $r\,(b-1) \ge \lambda + 128 = 256$ for Π_1. Part II tabulates (b, r) choices. (Purely interactive deployments need only $\kappa_1^r \le 2^{-\lambda}$, about half the repetitions.)

Remark 7.6 (Transcript compression). Under Fiat–Shamir the proof need not include (w, u): the verifier recomputes $w = (G_1\boldsymbol{\zeta}, G_2\boldsymbol{\zeta} + \mathbf{z}) - \gamma T_b$ and $u = (G_1\boldsymbol{\xi}, G_2\boldsymbol{\xi} + A\mathbf{z}) - \gamma T_c$ from the response, then checks that hashing these recomputed values reproduces γ. This is the classical Schnorr-style compression; it removes $2(k+n)\log q$ bits per repetition and makes the verification equations implicit in the hash check. The pseudocode in Section 12 uses it.

8 Protocol Π_2: All Three Matrices Hidden

8.1 What changes when A is hidden

Two structural shifts, before any protocol detail.

The statement becomes commitment-relative. With A hidden there is no longer an absolute anchor: the provable sentence is "the matrices sealed in T_A, T_B, T_C satisfy $C = AB$." Its value depends entirely on the *provenance* of T_A — why the verifier cares about that particular sealed A (a regulator holds it in escrow, it was published at model-release time, another proof connects it to signed data, etc.). The protocol proves consistency among containers; the application supplies meaning. We commit to A by *rows*: $T_A^{(i)} = \mathsf{com}(\mathbf{a}_i; \boldsymbol{\rho}_{A,i})$, $i \in [n]$, where $\mathbf{a}_i^\mathsf{T}$ is the i-th row — this orientation makes the left projection below homomorphic. As in Section 5.5, all statement commitments are well formed with $(\bar{T}_0, B_0)$.

The relation becomes bilinear. With A public, $B \mapsto AB$ is linear and Π_1's responses passed through it untouched. Now AB is a product of two secrets; linear masking no longer commutes with the relation, and some genuinely multiplicative step is unavoidable. The design principle is to make the bilinearity collapse to its minimum — one scalar product — before paying for it.

8.2 Double compression

Lemma 8.1 (Double compression). *Let $\mathbf{s} \leftarrow \mathcal{S}_b^n$, $\mathbf{v} \leftarrow \mathcal{S}_b^m$ be independent. If $C \neq AB \bmod q$ then*

$$\Pr\big[\,\mathbf{s}^\mathsf{T} C\,\mathbf{v} \;=\; \mathbf{s}^\mathsf{T} A\,B\mathbf{v} \pmod q\,\big] \;\le\; \frac{2}{2^b}.$$

Proof. Lemma 3.2 with $D = C - AB$. □

Define the three compressed secrets

$$\mathbf{a} := A^{\mathsf{T}}\mathbf{s} \in \mathbb{Z}_q^n, \qquad \mathbf{b} := B\mathbf{v} \in \mathbb{Z}_q^n, \qquad t := \mathbf{s}^{\mathsf{T}} C\, \mathbf{v} \in \mathbb{Z}_q,$$

so the claim to be proven is the single scalar equation $\langle \mathbf{a}, \mathbf{b} \rangle = t$. All three compressed secrets live in homomorphic reach of the statement commitments:

$$T_a := \sum_{i=1}^{n} s_i\, T_A^{(i)} = \mathsf{com}(\mathbf{a};\, \boldsymbol{\rho}_s), \quad T_b := \sum_{j=1}^{m} v_j\, T_B^{(j)} = \mathsf{com}(\mathbf{b};\, \boldsymbol{\rho}_v), \quad \boldsymbol{\rho}_s := \sum_i s_i \boldsymbol{\rho}_{A,i},\ \boldsymbol{\rho}_v := \sum_j v_j \boldsymbol{\rho}_j,$$

with $\|\boldsymbol{\rho}_s\| \le n2^b B_0 =: B_{\rho s}$ and $\|\boldsymbol{\rho}_v\| \le m2^b B_0 =: B_{\rho v}$. For the scalar t, project the *second slot* of the combined C-commitment by $\mathbf{s}$: writing $T_c = \sum_j v_j T_C^{(j)} = (G_1 \boldsymbol{\tau}_v,\ G_2 \boldsymbol{\tau}_v + C\mathbf{v})$, define

$$T_t := \big(T_{c,1},\ \ \mathbf{s}^{\mathsf{T}}\, T_{c,2}\big) = \big(G_1 \boldsymbol{\tau}_v,\ \ \mathbf{g}_s^{\mathsf{T}} \boldsymbol{\tau}_v + t\big), \qquad \mathbf{g}_s^{\mathsf{T}} := \mathbf{s}^{\mathsf{T}} G_2 \in \mathbb{Z}_q^{1\times N}.$$

That is, T_t is a commitment to the scalar t under the *derived scalar key* $(G_1, \mathbf{g}_s^{\mathsf{T}})$, with the same randomness $\boldsymbol{\tau}_v$ ($\|\boldsymbol{\tau}_v\| \le B_{\rho v}$). We write $\mathsf{com}_s(x; \boldsymbol{\rho}) = (G_1 \boldsymbol{\rho},\ \mathbf{g}_s^{\mathsf{T}} \boldsymbol{\rho} + x)$ for this key; its binding is inherited verbatim from G_1 (Lemma 5.5 used only the first slot and the primality of q), and both parties can form the key once $\mathbf{s}$ is public.

8.3 The product Σ-protocol

The one bilinear step. If the prover reveals masked versions $\mathbf{z}_a = \boldsymbol{\varepsilon}_a + \gamma \mathbf{a}$ and $\mathbf{z}_b = \boldsymbol{\varepsilon}_b + \gamma \mathbf{b}$, then

$$\langle \mathbf{z}_a, \mathbf{z}_b \rangle = \underbrace{\langle \boldsymbol{\varepsilon}_a, \boldsymbol{\varepsilon}_b \rangle}_{g_0} + \gamma\, \underbrace{\big(\langle \boldsymbol{\varepsilon}_a, \mathbf{b} \rangle + \langle \mathbf{a}, \boldsymbol{\varepsilon}_b \rangle\big)}_{g_1} + \gamma^2 \langle \mathbf{a}, \mathbf{b} \rangle \pmod q :$$

a quadratic in γ whose *leading coefficient is the claim* and whose lower coefficients are challenge-independent "garbage." Committing to g_0, g_1 *before* γ arrives turns the identity into a checkable equation with the claim isolated in the γ^2 slot. This is the classical product-proof pattern in its lattice form [18].

Protocol Π_2 (one repetition).

V1. $\mathcal{V}$ sends $\mathbf{s} \leftarrow \mathcal{S}_b^n$, $\mathbf{v} \leftarrow \mathcal{S}_b^m$.
Both parties form T_a, T_b, T_t and the scalar key $\mathbf{g}_s^{\mathsf{T}} = \mathbf{s}^{\mathsf{T}} G_2$ as above; $\mathcal{P}$ computes $\mathbf{a}, \mathbf{b}, \boldsymbol{\rho}_s, \boldsymbol{\rho}_v, \boldsymbol{\tau}_v$.

P1. $\mathcal{P}$ samples $\boldsymbol{\varepsilon}_a, \boldsymbol{\varepsilon}_b \leftarrow \mathbb{Z}_q^n$ uniform, $\boldsymbol{\eta}_1 \leftarrow \{-1, 0, 1\}^N$, and $(\boldsymbol{\eta}_a, \boldsymbol{\eta}_b, \boldsymbol{\eta}_0) \leftarrow \mathcal{D}_{\sigma_2}^{3N}$; computes the garbage scalars $g_0 = \langle \boldsymbol{\varepsilon}_a, \boldsymbol{\varepsilon}_b \rangle$, $g_1 = \langle \boldsymbol{\varepsilon}_a, \mathbf{b} \rangle + \langle \mathbf{a}, \boldsymbol{\varepsilon}_b \rangle \bmod q$; sends

$$w_a = \mathsf{com}(\boldsymbol{\varepsilon}_a; \boldsymbol{\eta}_a), \quad w_b = \mathsf{com}(\boldsymbol{\varepsilon}_b; \boldsymbol{\eta}_b), \quad t_0 = \mathsf{com}_s(g_0; \boldsymbol{\eta}_0), \quad t_1 = \mathsf{com}_s(g_1; \boldsymbol{\eta}_1).$$

V2. $\mathcal{V}$ sends $\gamma \leftarrow \mathcal{C} = \mathcal{S}_b$.

P2. $\mathcal{P}$ computes

$$\mathbf{z}_a = \boldsymbol{\varepsilon}_a + \gamma \mathbf{a}, \qquad \mathbf{z}_b = \boldsymbol{\varepsilon}_b + \gamma \mathbf{b} \pmod q;$$

$$\boldsymbol{\zeta}_a = \boldsymbol{\eta}_a + \gamma \boldsymbol{\rho}_s, \qquad \boldsymbol{\zeta}_b = \boldsymbol{\eta}_b + \gamma \boldsymbol{\rho}_v, \qquad \boldsymbol{\zeta}_q = \boldsymbol{\eta}_0 + \gamma \boldsymbol{\eta}_1 + \gamma^2 \boldsymbol{\tau}_v$$

(the last three over $\mathbb{Z}$), rejection-samples the concatenation $(\boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)$ against its center $(\gamma\boldsymbol{\rho}_s,\ \gamma\boldsymbol{\rho}_v,\ \gamma\boldsymbol{\eta}_1 + \gamma^2\boldsymbol{\tau}_v)$; on abort restarts from **P1** entirely fresh. On accept, sends $(\mathbf{z}_a, \mathbf{z}_b, \boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)$.

V3. $\mathcal{V}$ accepts iff $\|(\boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)\| \le B_2 := 1.2\,\sigma_2\sqrt{3N}$ and

$$\big(G_1\boldsymbol{\zeta}_a,\ G_2\boldsymbol{\zeta}_a + \mathbf{z}_a\big) = w_a + \gamma\, T_a \pmod q \qquad (\Pi_2\text{-i})$$

$$\big(G_1\boldsymbol{\zeta}_b,\ G_2\boldsymbol{\zeta}_b + \mathbf{z}_b\big) = w_b + \gamma\, T_b \pmod q \qquad (\Pi_2\text{-ii})$$

$$\big(G_1\boldsymbol{\zeta}_q,\ \mathbf{g}_s^{\mathsf{T}}\boldsymbol{\zeta}_q + \langle \mathbf{z}_a,\, \mathbf{z}_b\rangle\big) = t_0 + \gamma\, t_1 + \gamma^2\, T_t \pmod q. \qquad (\Pi_2\text{-iii})$$

Theorem 8.2 (Completeness of Π_2)**.** *If $C = AB \bmod q$, the honest prover convinces $\mathcal{V}$ with probability $1 - \mathsf{negl}(\lambda)$, restarting an expected $M \approx 2.76$ times. The required Gaussian width is $\sigma_2 = 12\,T_2$ with center bound*

$$T_2 = \Big(\|\gamma\boldsymbol{\rho}_s\|^2 + \|\gamma\boldsymbol{\rho}_v\|^2 + \|\gamma\boldsymbol{\eta}_1 + \gamma^2\boldsymbol{\tau}_v\|^2\Big)^{1/2}_{\max} \le \Big(2^{2b}(B_{\rho s}^2 + B_{\rho v}^2) + \big(2^b\sqrt{N} + 2^{2b}B_{\rho v}\big)^2\Big)^{1/2}.$$

Proof. Checks (Π_2-i), (Π_2-ii) are verbatim the Π_1 computation. For (Π_2-iii), first slot: $G_1\boldsymbol{\zeta}_q = G_1\boldsymbol{\eta}_0 + \gamma G_1\boldsymbol{\eta}_1 + \gamma^2 G_1\boldsymbol{\tau}_v$, matching $t_{0,1} + \gamma t_{1,1} + \gamma^2 T_{t,1}$. Second slot: by the displayed quadratic identity,

$$\mathbf{g}_s^{\mathsf{T}}\boldsymbol{\zeta}_q + \langle \mathbf{z}_a,\, \mathbf{z}_b\rangle = \big(\mathbf{g}_s^{\mathsf{T}}\boldsymbol{\eta}_0 + g_0\big) + \gamma\big(\mathbf{g}_s^{\mathsf{T}}\boldsymbol{\eta}_1 + g_1\big) + \gamma^2\big(\mathbf{g}_s^{\mathsf{T}}\boldsymbol{\tau}_v + \langle \mathbf{a},\, \mathbf{b}\rangle\big),$$

while the right side's γ^2 slot carries $\mathbf{g}_s^{\mathsf{T}}\boldsymbol{\tau}_v + t$. Equality holds iff $\langle \mathbf{a}, \mathbf{b}\rangle = t$, i.e. $\mathbf{s}^{\mathsf{T}}AB\mathbf{v} = \mathbf{s}^{\mathsf{T}}C\mathbf{v}$, true when $C = AB$. Norm and restart claims are Lemmas 6.4 and 6.3. □

8.4 Knowledge soundness: three transcripts

Theorem 8.3 (3-special soundness of Π_2)**.** *Fix $(\mathbf{s}, \mathbf{v})$ and three accepting transcripts with common first message (w_a, w_b, t_0, t_1) and pairwise distinct challenges $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{S}_b$. Assume $q > 2^{3b}$. Then either the extractor outputs a solution to $\mathsf{SIS}_{k,N,q,\beta_2}$ for G_1 with*

$$\beta_2 \;=\; \max\Big(\,\bar{T}_0 \cdot 2B_2 + 2^b B_{\rho s},\ \ \bar{T}_0 \cdot 3 \cdot 2^b B_2 + 2^{3b} B_{\rho v}\,\Big),$$

or the committed matrices satisfy $\mathbf{s}^{\mathsf{T}}(C - AB)\mathbf{v} = 0 \bmod q$. One repetition of Π_2 has knowledge error

$$\kappa_2 \;\le\; \frac{2}{2^b} + \frac{2}{2^b} + \mathsf{Adv}^{\mathsf{SIS}} \;=\; 2^{-(b-2)} + \mathsf{Adv}^{\mathsf{SIS}}.$$

Proof. Step 1: pin the linear parts. Any pair $i \ne j$ of transcripts, subtracted through (Π_2-i), yields (as in Theorem 7.2) a relaxed opening of T_a to the message $(\gamma_i - \gamma_j)^{-1}(\mathbf{z}_a^{(i)} - \mathbf{z}_a^{(j)})$ mod q with slack $< 2^b$ and norm $\le 2B_2$; comparing with the well-formed opening $(A^{\mathsf{T}}\mathbf{s}, \boldsymbol{\rho}_s, \bar{c}_0)$ via Lemma 5.5 either yields a SIS solution of norm $\le \bar{T}_0 2B_2 + 2^b B_{\rho s} \le \beta_2$ or forces

$$\mathbf{z}_a^{(i)} - \mathbf{z}_a^{(j)} = (\gamma_i - \gamma_j)\, A^{\mathsf{T}}\mathbf{s} \pmod q \quad \text{for all } i, j.$$

Hence $\mathbf{z}_a^{(i)} = \hat{\boldsymbol{\varepsilon}}_a + \gamma_i A^{\mathsf{T}}\mathbf{s}$ with $\hat{\boldsymbol{\varepsilon}}_a := \mathbf{z}_a^{(1)} - \gamma_1 A^{\mathsf{T}}\mathbf{s}$ fixed. Identically from (Π_2-ii): $\mathbf{z}_b^{(i)} = \hat{\boldsymbol{\varepsilon}}_b + \gamma_i B\mathbf{v}$. Therefore

$$\langle \mathbf{z}_a^{(i)}, \mathbf{z}_b^{(i)}\rangle \;=\; p(\gamma_i), \qquad p(X) := p_0 + p_1 X + p_2 X^2 \in \mathbb{Z}_q[X], \quad p_2 = \langle A^{\mathsf{T}}\mathbf{s},\, B\mathbf{v}\rangle = \mathbf{s}^{\mathsf{T}}AB\,\mathbf{v}.$$

Step 2: isolate the leading coefficient with integer weights. Set

$$\lambda_1 = \gamma_3 - \gamma_2, \quad \lambda_2 = \gamma_1 - \gamma_3, \quad \lambda_3 = \gamma_2 - \gamma_1,$$

so that $\sum_i \lambda_i = 0$, $\sum_i \lambda_i \gamma_i = 0$, and

$$\Delta := \sum_i \lambda_i \gamma_i^2 = -(\gamma_1 - \gamma_2)(\gamma_2 - \gamma_3)(\gamma_3 - \gamma_1) \neq 0, \qquad |\lambda_i| < 2^b, \quad 0 < |\Delta| < 2^{3b} < q.$$

(This is Lagrange interpolation for the leading coefficient with denominators cleared: $\sum_i \lambda_i\, p(\gamma_i) = p_2\, \Delta$ for every quadratic p, because the weights annihilate degrees 0 and 1.) Form the λ-weighted combination of the three instances of (Π_2-iii):

$$\Big(G_1 \textstyle\sum_i \lambda_i \boldsymbol{\zeta}_q^{(i)},\ \ \mathbf{g}_s^{\mathsf{T}} \sum_i \lambda_i \boldsymbol{\zeta}_q^{(i)} + p_2\, \Delta\Big) = \Delta\, T_t \pmod q,$$

the t_0, t_1 terms vanishing by the two annihilation identities. This exhibits $\big(p_2,\ \sum_i \lambda_i \boldsymbol{\zeta}_q^{(i)},\ \Delta\big)$ as a relaxed opening of T_t with slack $< 2^{3b}$ and norm $\le 3 \cdot 2^b B_2$.

Step 3: bind. The well-formed chain gives the relaxed opening $(t, \boldsymbol{\tau}_v, \bar{c}_0)$ of T_t under the same key, with $t = \mathbf{s}^{\mathsf{T}} C \mathbf{v}$ and norm $\le B_{\rho v}$. Lemma 5.5 (first slot only involves G_1, so it applies to the scalar key unchanged) either yields a SIS solution of norm $\le \bar{T}_0 \cdot 3 \cdot 2^b B_2 + 2^{3b} B_{\rho v} \le \beta_2$, or forces $p_2 = t$, i.e. $\mathbf{s}^{\mathsf{T}} AB \mathbf{v} = \mathbf{s}^{\mathsf{T}} C \mathbf{v} \bmod q$.

Knowledge error. Condition on no SIS break. If $C \neq AB$, then by Lemma 8.1 the compression misses with probability $\le 2 \cdot 2^{-b}$; otherwise no first message answers three distinct challenges, so by the 3-special-soundness rewinding bound the prover survives with probability $\le 2/|\mathcal{C}| = 2 \cdot 2^{-b}$ [19]. □

Remark 8.4 (Why 2^{3b} appears, and its price). The Vandermonde clearing multiplies the slack by a product of *three* challenge differences; that 2^{3b} lands inside β_2 and (via $q > \beta_2$ headroom) inside $\log q$. This is the concrete reason Part II selects a smaller b for Π_2 than for Π_1, taking more repetitions in exchange.

8.5 Zero knowledge

Theorem 8.5 (HVZK of Π_2)**.** *There is a simulator whose transcripts are within 2^{-99} statistical distance of honest ones if the commitment is statistically hiding, and computationally indistinguishable under* kLWE *in the computational mode.*

Proof. Simulator: $\gamma \leftarrow \mathcal{C}$; $\mathbf{z}_a, \mathbf{z}_b \leftarrow \mathbb{Z}_q^n$ uniform; $(\boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q) \leftarrow \mathcal{D}_{\sigma_2}^{3N}$ conditioned on the norm check; $t_1 := \mathsf{com}_s(0; \boldsymbol{\eta}_1')$ for fresh ternary $\boldsymbol{\eta}_1'$; then back-solve the three verification equations for the remaining first-message parts:

$$w_a := (G_1 \boldsymbol{\zeta}_a, G_2 \boldsymbol{\zeta}_a + \mathbf{z}_a) - \gamma T_a, \quad w_b := (\cdots) - \gamma T_b, \quad t_0 := \big(G_1 \boldsymbol{\zeta}_q,\ \mathbf{g}_s^{\mathsf{T}} \boldsymbol{\zeta}_q + \langle \mathbf{z}_a,\, \mathbf{z}_b \rangle\big) - \gamma t_1 - \gamma^2 T_t.$$

As in Theorem 7.4, the marginals of $(\gamma, \mathbf{z}_a, \mathbf{z}_b, \boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)$ match the real ones statistically (Lemmas 6.1, 6.3), and w_a, w_b, t_0 are the same deterministic functions of them in both worlds. The one residual discrepancy is t_1: real t_1 commits to $g_1 \neq 0$ in general, simulated t_1 commits to 0. Commitment hiding (Lemma 5.6 or 5.7) makes the two t_1-distributions indistinguishable in the respective mode, and a hybrid over this single component finishes. □

Composition (parallel repetition, Fiat–Shamir, grinding budget) is exactly as in Section 7.5, with per-repetition error $\kappa_2 \approx 2^{-(b-2)}$ and target $r(b-2) \ge 256$.

9 Concrete Security: the Cost of Every Attack

This section prices each avenue of attack under the standard concrete-hardness methodology used by the NIST post-quantum process [13, 26, 25]. The numbers here justify the parameter targets of Part II.

9.1 The attack surface, enumerated

Goal	What it requires	Cost at our parameters
Recover B, C (or A) from commitments/transcripts	Statistical mode: beat a 2^{-128} statistical distance — no computation helps. Computational mode: solve LWE of dimension $N-(k+n)$	Mode i: information-theoretically excluded. Mode ii: $\geq 2^{128}$ ops (Section 9.2)
Forge: convince verifier of $C \neq AB$	Produce a short nonzero kernel vector of G_1 ($\mathsf{SIS}_{k,N,q,\beta}$), by Theorems 7.2/8.3	$\geq 2^{128}$ ops
Get lucky per proof attempt	Hit the κ^r event, amplified by hash grinding	$\leq 2^{-128}$ after the $(Q{+}1)^2$ budget of Section 7.5

9.2 Lattice attack cost model

The best known algorithms for both SIS and LWE reduce to lattice basis reduction with the BKZ algorithm, whose dominant subroutine is solving the shortest-vector problem in a projected block of dimension β_{blk}. The *core-SVP* methodology conservatively prices an attack at the cost of a *single* such SVP call, itself performed by the asymptotically best sieve [14]:

$$\text{classical: } 2^{0.292\,\beta_{\mathrm{blk}}}, \qquad \text{quantum: } \approx 2^{0.26\,\beta_{\mathrm{blk}}}, \qquad \text{memory: } \approx 2^{0.208\,\beta_{\mathrm{blk}}}.$$

For a q-ary SIS lattice, BKZ with root Hermite factor δ_0 finds vectors of norm about $\min_{d \leq N} \delta_0^d\, q^{k/d} = 2^{2\sqrt{k \log_2 q \, \log_2 \delta_0}}$ (the optimum over the sublattice dimension d). Requiring this to exceed the largest norm β our extraction ever needs gives the design inequality

$$2\sqrt{k \cdot \log_2 q \cdot \log_2 \delta_0(\beta_{\mathrm{blk}})} \;\geq\; \log_2 \beta, \qquad 0.292\,\beta_{\mathrm{blk}} \geq 128 \;\Rightarrow\; \beta_{\mathrm{blk}} \geq 439, \quad \delta_0 \approx 1.0037. \quad (\star)$$

Three sanity anchors for what 2^{128} classical operations means: an exascale machine (10^{18} ops/s) needs about 10^{13} years; the entire Bitcoin network's hash throughput, repurposed, needs on the order of the age of the universe; and the sieve's $2^{0.208 \cdot 439} \approx 2^{91}$-bit memory footprint exceeds global storage manufactured to date by orders of magnitude — the memory wall is often the more binding constraint in practice, and core-SVP ignores it, which is a conservatism in our favor.

9.3 Quantum computers

Shor's algorithm, which breaks factoring- and discrete-log-based systems outright, has no known analogue for SIS/LWE; lattice problems are the basis of the NIST post-quantum standards precisely for this reason. Known quantum speedups for sieving are Grover-flavored, improving the exponent from 0.292 to roughly 0.26 [26]: block 439 still costs $\approx 2^{114}$ quantum operations, each vastly more expensive than a classical one on any foreseeable hardware, with unresolved QRAM requirements. Deployments wanting $\geq 2^{128}$ under the quantum exponent should size ($\star$) with $\beta_{\mathrm{blk}} \geq 500$ ($\delta_0 \approx 1.0033$); the parameter machinery of Part II accepts this as a one-line change.

9.4 Grinding, multi-proof, and quantum access to the oracle

Fiat–Shamir grinding is already budgeted in Section 7.5 at $Q = 2^{64}$ queries with a $(Q+1)^2$ multi-round loss — a deliberately blunt, safe accounting. Issuing many proofs against the same commitments does not erode secrecy: each transcript is simulatable (Theorems 7.4, 8.5), and statistical distances add linearly, so even 2^{40} proofs stay below 2^{-59} total transcript leakage in the statistical mode. Against quantum adversaries with superposition access to the hash, the measure-and-reprogram line of results supports Fiat–Shamir for (multi-round) special-sound protocols with comparable losses; we cite this as out-of-scope engineering margin rather than re-deriving it [20].

9.5 What breaking each assumption would mean

SIS and kLWE/LWE at these parameters underpin the NIST standards ML-KEM (Kyber) and ML-DSA (Dilithium) [26, 25]. An adversary breaking the binding or (computational-mode) hiding of this protocol at 2^{128} scale would, with the same machinery, break the encryption and signatures now being deployed across operating systems, browsers, and national infrastructure. The protocol's security is therefore not a bespoke claim: it rides the most heavily scrutinized hardness frontier in post-quantum cryptography, with Ajtai's worst-case reduction [6] standing behind the average case.

Part II
Implementation

10 Instantiation Routes

Two ways to realize the algebra of Part I in software.

Route Z (plain $\mathbb{Z}_q$; the reference implementation). Exactly the objects of Part I: integer vectors modulo a prime q of at most two machine words, scalar challenges from $\mathcal{S}_b$, security assembled by r parallel repetitions. Every kernel is a matrix–vector or matrix–matrix product over $\mathbb{Z}_q$. This route maximizes transparency (the code is a line-by-line transcription of the protocol boxes), minimizes novel implementation risk, and is the normative reference against which any optimized implementation should be tested. Its costs — worked out below — are proofs in the megabytes and prover time in seconds; entirely usable for high-value/low-frequency proofs, wasteful for high-frequency ones.

Route R (module lattices over a cyclotomic ring). Replace $\mathbb{Z}_q$ by $R_q = \mathbb{Z}_q[X]/(X^d+1)$ with $d \in \{64, 128\}$. One ring element packs d scalars; NTT-based multiplication turns the big matrix kernels into batched $d \log d$ transforms; and, decisively, the challenge set can be the exponentially large set of *small-coefficient polynomials* (e.g. ternary coefficients: $3^{128} \approx 2^{202}$ elements of unit-scale norm), so *one* round replaces all r repetitions and proofs shrink by one to two orders of magnitude. The costs are conceptual: Schwartz–Zippel and challenge-difference invertibility need care in a ring with zero divisors (resolved by choosing q so $X^d + 1$ splits into few factors and invoking the short-element invertibility theorem of [16]), and one genuinely structural caveat discussed in Section 17 — the public-A protocol Π_1 uses arbitrary $\mathbb{Z}_q$-linear maps, which do not commute with ring multiplication unless A is itself expressed over the ring. Production systems in this family [15, 18, 21] all take Route R.

This document specifies Route Z normatively (Sections 11–16) and maps the lift to Route R in Section 17. Build Z first; validate; then port hot paths to R with Z as the differential-testing oracle.

11 Parameter Selection

11.1 The constraint system

Free inputs: matrix sizes (n, m); security $\lambda = 128$; hash-grinding budget $Q = 2^{64}$; coefficient/challenge width b; repetitions r; randomness dimension N; binding rows k; modulus q. Derived quantities, for $B_0 = \sqrt{N}$ (ternary statement randomness), tail factor 1.2, rejection ratio $\alpha = 12$:

	Π_1	Π_2
combined-randomness norms	$B_\rho = m2^b\sqrt{N}$	$B_{\rho s} = n2^b\sqrt{N},\ B_{\rho v} = m2^b\sqrt{N}$
rejection center bound	$T_1 = \sqrt{2}\,2^b B_\rho$	$T_2 \approx \left(2^{2b}(B_{\rho s}^2+B_{\rho v}^2) + (2^b\sqrt{N}+2^{2b}B_{\rho v})^2\right)^{1/2}$
Gaussian width / bound	$\sigma_1 = 12T_1,\ B_1 = 1.2\sigma_1\sqrt{2N}$	$\sigma_2 = 12T_2,\ B_2 = 1.2\sigma_2\sqrt{3N}$
extraction (SIS) norm	$\beta_1 = 2B_1 + 2^b B_\rho$	$\beta_2 = \max(2B_2+2^b B_{\rho s},\ 3\cdot 2^b B_2+2^{3b}B_{\rho v})$

Constraints (each names the theorem it discharges):

C0 q prime, $q > 2^{3b}$ (nonzero slack mod q; Theorem 8.3)

C1 $r(b-1) \geq \lambda + 2\log_2 Q = 256$ for Π_1; $r(b-2) \geq 256$ for Π_2 (Section 7.5)

C2 $q > 8\,\beta$ (SIS meaningful; headroom below the trivial q-vector)

C3 $2\sqrt{k\,\log_2 q\,\log_2 \delta_0} \geq \log_2 \beta$ with $\delta_0 = 1.0037$ (binding $\geq 2^{128}$; eq. $(\star)$)

C4 computational hiding: $N - (k+n) \geq d_{\mathsf{LWE}}(q)$, with the crude first-cut scaling $d_{\mathsf{LWE}}(q) \approx 768 \cdot \log_2 q/11.7$ (Kyber-normalized ternary-noise scaling) (Lemma 5.7)

C4′ statistical hiding alternative: $N\log_2 3 \geq (k+n)\log_2 q + 258$ (Lemma 5.6)

C5 integer-semantics option: $q > 2n \cdot 2^{a+b_B}$ (Remark 2.1)

Status of the numbers below. C3 and C4 are back-of-envelope models (core-SVP with a fixed δ_0; a linear LWE-dimension scaling). They are the right shape for design iteration but are *not* a security review. Before deployment, validate every candidate set with the lattice estimator of Albrecht–Player–Scott [13] (maintained as `malb/lattice-estimator`); Section 16 gives the invocation. Expect the estimator to move k and N by tens of percent in either direction; the constraint structure, not the specific integers, is the deliverable of this section.

11.2 Worked illustrative sets ($n = m = 1024$)

Solving the system by iteration (choose b, r from C1; guess N; evaluate norms; size q by C2, k by C3; check C4; re-iterate once) yields:

	Π_1, set A	Π_1, set B	Π_2
b (coefficient/challenge bits)	24	8	12
r (repetitions)	12	37	26
$\lceil \log_2 q \rceil$	88 (two limbs)	54 (**one limb**)	88 (two limbs)
N (randomness dim.)	10240	6656	10240
k (binding rows)	3200	1728	3200
$\log_2 \sigma$	68.8	36.4	56.3
$\log_2 B$ (response bound)	76.2	43.5	64.0
$\log_2 \beta$ (SIS norm)	77.2	44.6	77.6
commitment / column (KB)	46.5	18.6	46.5
proof size, total (MB)	≈ 2.4	≈ 2.7	≈ 6.6

Sample derivation (set A, so the arithmetic is reproducible): $B_\rho = 2^{10} \cdot 2^{24} \cdot \sqrt{10240} = 2^{40.7}$; $T_1 = \sqrt{2} \cdot 2^{24} \cdot 2^{40.7} = 2^{65.2}$; $\sigma_1 = 12\, T_1 = 2^{68.8}$; $B_1 = 1.2 \cdot 2^{68.8} \cdot \sqrt{20480} = 2^{76.2}$; $\beta_1 = 2^{77.2}$; C2 gives $q \approx 2^{88}$; C3 gives $k \geq (77.2/2)^2/(88 \cdot 0.00537) \approx 3150 \to 3200$; C4 needs $N \geq 3200 + 1024 + 768 \cdot 88/11.7 \approx 10000 \to N = 10240$; one re-iteration confirms. Set B trades 4.6× more repetitions for single-word arithmetic — usually the better engineering deal. The Π_2 column shows the 2^{3b} slack of Theorem 8.3 at work: b drops to 12 and r rises to 26 to keep q at two limbs.

For the statistical-hiding mode, C4′ replaces C4: set B would need $N \geq (2752 \cdot 54 + 258)/1.585 \approx 94{,}000$ — a 14× dimension surcharge, the concrete price of Corollary 4.4's unconditional option.

11.3 Knobs and their gradients

Raising b: fewer repetitions (C1) but +1 bit on every norm per unit of b for Π_1 and +3 bits on the β_2 slack for Π_2; the optimum sits where $\log q$ crosses a machine-word boundary. Raising N: helps C4 linearly, hurts norms only through $\sqrt{N}$; the cheapest security dial. Raising k: helps C3 as $\sqrt{k}$, costs commitment size linearly. Larger matrices (n, m): enter norms only logarithmically ($B_\rho \propto m$), so parameters are insensitive to problem scale — the design amortizes well as matrices grow.

12 Algorithms

Complete pseudocode for Route Z. Conventions: H is SHAKE-256 used as an extendable-output function; every call takes an explicit ASCII domain-separation tag as its first argument; $\mathsf{crs_hash} = H(\texttt{"crs"}, \text{seed}, q, n, m, k, N, b, r)$ is bound into every transcript hash. All mod-q elements travel in canonical representatives; all short vectors travel as signed integers.

Discrete Gaussians $\mathcal{D}_\sigma$: for these very wide σ (2^{36}–2^{69}), sample per coordinate as (uniform integer in a $\pm 12\sigma$ window filtered by the Gaussian density) or via Karney's exact method; constant-time behavior is discussed in Section 15. The exponential in Alg. 2 must be evaluated in fixed point to ≥ 128 fractional bits (the quantities are ratios of huge integers; double precision silently biases the output — a known implementation trap).

The Π_2 verifier mirrors Alg. 4: recompute w_a, w_b from (Π_2-i), (Π_2-ii), recompute $t_0 = (G_1 \boldsymbol{\zeta}_q,\ \mathbf{g}_s^{\mathsf{T}} \boldsymbol{\zeta}_q + \langle \mathbf{z}_a, \mathbf{z}_b \rangle) - \gamma t_1 - \gamma^2 T_t$, check the norm bound B_2, membership $\gamma \in \mathcal{S}_b$, and the hash equation. Note t_1 *is* transmitted (it cannot be recomputed — it is the one first-message component

Algorithm 1 Setup and commitment primitives

procedure ExpandCRS(seed)
  stream $\leftarrow H(\texttt{"G1G2"}, \text{seed})$
  fill $G_1 \in \mathbb{Z}_q^{k\times N}$, then $G_2 \in \mathbb{Z}_q^{n\times N}$, reading $\lceil \log_2 q \rceil$-bit chunks and rejecting values $\geq q$ ▷ unbiased
  return (G_1, G_2)
end procedure
procedure Commit($\mathbf{x} \in \mathbb{Z}_q^n$; $\boldsymbol{\rho} \in \{-1, 0, 1\}^N$)
  return $t = (G_1\boldsymbol{\rho} \bmod q,\ \ G_2\boldsymbol{\rho} + \mathbf{x} \bmod q)$
end procedure
procedure CommitMatrix(B) ▷ columns; rows for T_A via $B \leftarrow A^{\mathsf{T}}$
  for $j = 1 \dots m$ **do** $\boldsymbol{\rho}_j \leftarrow$ ternary from $H(\texttt{"rand"}, \text{sk-seed}, j)$; $\ T^{(j)} \leftarrow \text{Commit}(\mathbf{b}_j; \boldsymbol{\rho}_j)$
  end for
end procedure
procedure ProveOpening($t, \boldsymbol{\rho}$) ▷ well-formedness, Section 5.5
  $\mathbf{y} \leftarrow \mathcal{D}_{\sigma_o}^N$; $\ w \leftarrow G_1\mathbf{y} \bmod q$; $\ \gamma \leftarrow H(\texttt{"open"}, \mathsf{crs_hash}, t, w) \in \mathcal{S}_b$
  $\boldsymbol{\zeta} \leftarrow \mathbf{y} + \gamma\boldsymbol{\rho}$; rejection-sample (Alg. 2); on abort restart
  return $(w, \boldsymbol{\zeta})$; $\ $ verifier checks $G_1\boldsymbol{\zeta} = w + \gamma\, t_1$ and $\|\boldsymbol{\zeta}\| \leq 1.2\,\sigma_o\sqrt{N}$
end procedure

Algorithm 2 Rejection sampling (Lemma 6.3); $\alpha = 12$, $M = e^{1+1/288}$

procedure RejSample($\boldsymbol{\zeta}, \mathbf{c}, \sigma$) ▷ $\boldsymbol{\zeta} = \mathbf{y} + \mathbf{c}$ already formed
  $u \leftarrow [0, 1)$ uniform from $H(\texttt{"rej"}, \dots)$
  return accept iff $u < \dfrac{1}{M}\exp\!\left(\dfrac{\|\mathbf{c}\|^2 - 2\langle\boldsymbol{\zeta}, \mathbf{c}\rangle}{2\sigma^2}\right)$ ▷ clamped to 1
end procedure

Algorithm 3 Π_1: non-interactive prover (Fiat–Shamir, r repetitions)

procedure Prove1($A, B, C, \{\boldsymbol{\rho}_j, \boldsymbol{\tau}_j\}, \{T_B^{(j)}, T_C^{(j)}\}$)
  $h_0 \leftarrow H(\texttt{"stmt"}, \mathsf{crs_hash}, A, \{T_B^{(j)}, T_C^{(j)}\})$
  for $\ell = 1 \dots r$ **do**
    $\mathbf{v}^{(\ell)} \leftarrow H(\texttt{"v"}, h_0, \ell) \in \mathcal{S}_b^m$; $\ \mathbf{b} \leftarrow B\mathbf{v}^{(\ell)}$; $\boldsymbol{\rho}_v \leftarrow \sum_j v_j\boldsymbol{\rho}_j$; $\boldsymbol{\tau}_v \leftarrow \sum_j v_j\boldsymbol{\tau}_j$
    repeat
      $\boldsymbol{\varepsilon} \leftarrow \mathbb{Z}_q^n$; $(\boldsymbol{\eta}, \boldsymbol{\mu}) \leftarrow \mathcal{D}_{\sigma_1}^{2N}$ ▷ all fresh on every abort
      $w \leftarrow \text{Commit}(\boldsymbol{\varepsilon}; \boldsymbol{\eta})$; $\ u \leftarrow \text{Commit}(A\boldsymbol{\varepsilon}; \boldsymbol{\mu})$
      $\gamma^{(\ell)} \leftarrow H(\texttt{"gamma"}, h_0, \ell, w, u) \in \mathcal{S}_b$
      $\mathbf{z} \leftarrow \boldsymbol{\varepsilon} + \gamma^{(\ell)}\mathbf{b} \bmod q$; $\ (\boldsymbol{\zeta}, \boldsymbol{\xi}) \leftarrow (\boldsymbol{\eta}, \boldsymbol{\mu}) + \gamma^{(\ell)}(\boldsymbol{\rho}_v, \boldsymbol{\tau}_v)$
    until RejSample($(\boldsymbol{\zeta}, \boldsymbol{\xi}),\ \gamma^{(\ell)}(\boldsymbol{\rho}_v, \boldsymbol{\tau}_v),\ \sigma_1$)
    record $\pi_\ell \leftarrow (\gamma^{(\ell)}, \mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$ ▷ w, u omitted; Remark 7.6
  end for
  return $\pi = (\pi_1, \dots, \pi_r)$
end procedure

Algorithm 4 Π_1: verifier

procedure VERIFY1($A, \{T_B^{(j)}, T_C^{(j)}\}, \pi$)
  $h_0 \leftarrow H(\texttt{"stmt"}, \dots)$ as in Alg. 3
  for $\ell = 1 \dots r$ **do**
    parse $(\gamma^{(\ell)}, \mathbf{z}, \boldsymbol{\zeta}, \boldsymbol{\xi})$; **require** $\|(\boldsymbol{\zeta}, \boldsymbol{\xi})\| \le B_1$ **and** $\gamma^{(\ell)} \in \mathcal{S}_b$
    $\mathbf{v}^{(\ell)} \leftarrow H(\texttt{"v"}, h_0, \ell)$; $T_b \leftarrow \sum_j v_j T_B^{(j)}$; $T_c \leftarrow \sum_j v_j T_C^{(j)}$
    $w \leftarrow (G_1\boldsymbol{\zeta},\ G_2\boldsymbol{\zeta} + \mathbf{z}) - \gamma^{(\ell)} T_b \bmod q$
    $u \leftarrow (G_1\boldsymbol{\xi},\ G_2\boldsymbol{\xi} + A\mathbf{z}) - \gamma^{(\ell)} T_c \bmod q$
    require $\gamma^{(\ell)} = H(\texttt{"gamma"}, h_0, \ell, w, u)$
  end for
  return accept
end procedure

Algorithm 5 Π_2: non-interactive prover (one repetition shown inside the same r-loop and abort loop as Alg. 3)

$(\mathbf{s}, \mathbf{v}) \leftarrow H(\texttt{"sv"}, h_0, \ell) \in \mathcal{S}_b^n \times \mathcal{S}_b^m$; $\mathbf{a} \leftarrow A^{\mathsf{T}}\mathbf{s}$; $\mathbf{b} \leftarrow B\mathbf{v}$; $\mathbf{g}_s^{\mathsf{T}} \leftarrow \mathbf{s}^{\mathsf{T}} G_2$
$\boldsymbol{\rho}_s \leftarrow \sum_i s_i \boldsymbol{\rho}_{A,i}$; $\boldsymbol{\rho}_v \leftarrow \sum_j v_j \boldsymbol{\rho}_j$; $\boldsymbol{\tau}_v \leftarrow \sum_j v_j \boldsymbol{\tau}_j$
$\boldsymbol{\varepsilon}_a, \boldsymbol{\varepsilon}_b \leftarrow \mathbb{Z}_q^n$; $\boldsymbol{\eta}_1 \leftarrow \{-1, 0, 1\}^N$; $(\boldsymbol{\eta}_a, \boldsymbol{\eta}_b, \boldsymbol{\eta}_0) \leftarrow \mathcal{D}_{\sigma_2}^{3N}$
$g_0 \leftarrow \langle \boldsymbol{\varepsilon}_a,\ \boldsymbol{\varepsilon}_b \rangle$; $g_1 \leftarrow \langle \boldsymbol{\varepsilon}_a,\ \mathbf{b} \rangle + \langle \mathbf{a},\ \boldsymbol{\varepsilon}_b \rangle \bmod q$
$w_a \leftarrow \text{COMMIT}(\boldsymbol{\varepsilon}_a; \boldsymbol{\eta}_a)$; $w_b \leftarrow \text{COMMIT}(\boldsymbol{\varepsilon}_b; \boldsymbol{\eta}_b)$; $t_0 \leftarrow \mathsf{com}_s(g_0; \boldsymbol{\eta}_0)$; $t_1 \leftarrow \mathsf{com}_s(g_1; \boldsymbol{\eta}_1)$
$\gamma \leftarrow H(\texttt{"gamma2"}, h_0, \ell, w_a, w_b, t_0, t_1) \in \mathcal{S}_b$
$\mathbf{z}_a \leftarrow \boldsymbol{\varepsilon}_a + \gamma\mathbf{a}$; $\mathbf{z}_b \leftarrow \boldsymbol{\varepsilon}_b + \gamma\mathbf{b} \pmod q$
$\boldsymbol{\zeta}_a \leftarrow \boldsymbol{\eta}_a + \gamma\boldsymbol{\rho}_s$; $\boldsymbol{\zeta}_b \leftarrow \boldsymbol{\eta}_b + \gamma\boldsymbol{\rho}_v$; $\boldsymbol{\zeta}_q \leftarrow \boldsymbol{\eta}_0 + \gamma\boldsymbol{\eta}_1 + \gamma^2\boldsymbol{\tau}_v$
rejection-sample $(\boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)$ against center $(\gamma\boldsymbol{\rho}_s, \gamma\boldsymbol{\rho}_v, \gamma\boldsymbol{\eta}_1 + \gamma^2\boldsymbol{\tau}_v)$; abort $\Rightarrow$ restart repetition with all-fresh randomness
emit $\pi_\ell = (\gamma, t_1, \mathbf{z}_a, \mathbf{z}_b, \boldsymbol{\zeta}_a, \boldsymbol{\zeta}_b, \boldsymbol{\zeta}_q)$ ▷ w_a, w_b, t_0 recomputed by verifier

not determined by the response; compare Theorem 8.5, where it is also the one component whose simulation consumes hiding).

13 Arithmetic and Performance Engineering

Modular arithmetic. Set B ($q < 2^{54}$) fits single-word Montgomery or Barrett arithmetic with lazy reduction — the entire prover is then native 64-bit integer code. Sets A and the Π_2 set ($q \approx 2^{88}$) want two-limb Montgomery; on x86-64, `mulx/adcx/adox` chains, on Apple/ARM, `umulh`. Choose $q \equiv 1 \pmod{2^{20}}$-style primes only if the Route-R port is planned (NTT friendliness); Route Z needs no structure in q beyond primality and C0/C2.

Kernel shape. Everything expensive is a product against G_1, G_2. Batch aggressively: the r repetitions' commitments $(\boldsymbol{\eta}^{(\ell)})_\ell$ form an $N \times 2r$ matrix, so committing is one BLAS-3-shaped $(k{+}n) \times N$ by $N \times 2r$ product — cache-blocked, vectorized (AVX-512 IFMA does 52-bit lanes that match set B beautifully), and embarrassingly GPU-friendly (these are exactly the GEMM shapes GPUs are built for). The homomorphic combinations $\sum_j v_j T^{(j)}$ are $(k{+}n) \times m$ by $m \times r$ products with small v-entries — use the b-bit width: for set B these are 8-bit multiplies.

Memory and storage. G_1, G_2 need never be stored: expand from seed in streamed panels during each GEMM pass (SHAKE at ∼GB/s is not the bottleneck against memory bandwidth). Statement commitments are the dominant persistent object ($(k{+}n)\lceil \log q \rceil$ per column; tens of MB per 1024-column matrix at the worked parameters); store column-major, memory-map for verification.

Wall-clock shape (set B, one modern 16-core CPU, order-of-magnitude). One-time commitment to B and C: $\sim 10^{11}$ word-multiplies, a few seconds multithreaded, sub-second on a GPU; amortized across every proof made against those commitments. Per proof: $r \cdot M \cdot 2(k{+}n)N \approx 4 \times 10^9$ word-multiplies plus one $n \times n$ matvec per repetition — roughly a second. Verification: similar shape, no rejection factor M, slightly cheaper. Proof sizes as tabulated in Section 11.2.

14 Cost Accounting

Symbolic operation counts (mod-q multiplies; $M \approx 2.76$):

Phase	Π_1	Π_2
Commit (one-time)	$2m(k{+}n)N$	$(2m{+}n)(k{+}n)N$
Prove, per proof	$r\big[\,2M(k{+}n)N + m(k{+}n) + n^2\,\big]$	$r\big[\,M(2(k{+}n){+}2)N + (n{+}m)(k{+}n) + 2n^2\,\big]$
Verify, per proof	$r\big[\,2(k{+}n)N + m(k{+}n) + n^2\,\big]$	$r\big[\,(2(k{+}n){+}2)N + (n{+}m)(k{+}n) + 2n^2\,\big]$
Proof size (bits)	$r\big[n \log q + 2N(\log \sigma_1{+}4)\big]$	$r\big[(2n{+}k{+}1)\log q + 3N(\log \sigma_2{+}4)\big]$

Two honest ratios to internalize. First, committing costs 25–100× the bare n^2m multiply count of the matrix product itself at the worked parameters — the price of Route Z's transparency, paid once per matrix and amortized over all proofs; Route R compresses this toward small single-digit factors. Second, proofs are megabytes, not kilobytes: linear in $r \cdot N$. Route R's single repetition and packed ring elements bring this to the 50–150 KB range typical of the literature [18, 21], and the folding systems of Section 17 go sublinear.

15 Implementation Security

Checklist — each item is a known, exploited failure class.

1. *Fresh randomness on every abort.* On rejection, resample *everything* ($\boldsymbol{\varepsilon}$-type and $\boldsymbol{\eta}$-type) — reusing the message mask across attempts voids Lemma 6.3's independence guarantee. One mask, one challenge, ever; mask reuse is the analogue of ECDSA nonce reuse and yields secret-recovery by differencing.
2. *Verify every norm bound and set membership.* Omitting the $\|\cdot\| \le B$ check, or the $\gamma \in \mathcal{S}_b$ / decoded-challenge range checks, silently removes the SIS reduction — the classic "forgot the norm check" lattice bug. Check norms in exact integer arithmetic.
3. *Hash everything, domain-separate everything.* Every challenge hash must bind $\mathsf{crs_hash}$, the full statement, the repetition index, and all prior messages. Omission of the statement enables cross-statement proof transplantation.
4. *Constant-time secret handling.* Gaussian sampling and the rejection decision must not branch on secret values in observable time. Under Fiat–Shamir the *number* of aborts is not in the transcript, but co-located attackers can time it; Karney/CDT samplers with constant-time comparisons are the standard remedy [25]. Zeroize masks and combined randomness after use.
5. *Exact rejection arithmetic.* The accept probability involves exp of ratios of $\sim 2^{140}$-scale integers; evaluate in fixed point with ≥ 128 fractional bits (Section 12).
6. *No modulus ambiguity.* Fix canonical representatives once; a verifier accepting both representative conventions for $\mathbf{z}$ effectively doubles the response space and breaks the extraction argument's uniqueness bookkeeping.

16 Testing and Validation

Functional. Known-answer tests for ExpandCRS, Commit, homomorphism identities ($\mathsf{com}(\mathbf{x};\boldsymbol{\rho}) + c\,\mathsf{com}(\mathbf{x}';\boldsymbol{\rho}') = \mathsf{com}(\mathbf{x}+c\mathbf{x}';\boldsymbol{\rho}+c\boldsymbol{\rho}')$ for random small c); end-to-end completeness on random true instances at reduced dimensions (e.g. $n = m = 8$, $N = 64$, toy q) and at full parameters.

Adversarial. Each test must *fail* verification: flip one entry of C after committing; replace one $T_C^{(j)}$ with a fresh commitment to the same column (hash binding must catch it); tamper with one coordinate of $\mathbf{z}$, of $\boldsymbol{\zeta}$, of γ; submit a response with norm just above B; replay a repetition's transcript under another repetition index; swap statements under one proof.

Statistical (zero-knowledge health). Over $\ge 10^5$ transcripts at toy parameters: chi-square uniformity of $\mathbf{z}$ coordinates; Kolmogorov–Smirnov of $\boldsymbol{\zeta}$ marginals against $\mathcal{D}_\sigma$; empirical correlation of $\boldsymbol{\zeta}$ against the secret direction $\boldsymbol{\rho}_v$ (must be statistically zero — this is the test that catches rejection-sampling bugs); abort-rate confidence interval around $1 - 1/M$.

Hardness validation. Run the lattice estimator [13] on the final parameters; schematically:

```
from estimator import *
SIS.estimate(SIS.Parameters(n=k, q=q, length_bound=beta, m=N))
LWE.estimate(LWE.Parameters(n=N-(k+n), q=q, Xs=ternary, Xe=ternary))
```

and require every reported attack $\ge 2^{128}$ (or $\ge 2^{128}$ quantum, per the chosen threat model), then re-run after any parameter change, however minor.

17 Extensions and the Path to Production

The ring lift (Route R). Take $R_q = \mathbb{Z}_q[X]/(X^d+1)$, $d = 128$, with q chosen so X^d+1 factors into two irreducible halves ($q \equiv 5 \bmod 8$); then every nonzero $y \in R_q$ with $\|y\|_\infty < q^{1/2}/\sqrt{2}$ is invertible [16], which restores the two facts Part I used primality for: Freivalds-type compression (per-CRT-slot Schwartz–Zippel) and nonzero challenge differences. Challenges become ternary-coefficient ring elements (3^{128} of them at unit norm): $r = 1$, norms shrink by the former 2^b factors, q drops to one 32–64-bit limb, and all kernels become NTT-batched. One structural caveat: Π_1's check (Π_1-ii) applies the arbitrary $\mathbb{Z}_q$-linear map A to a response; over R_q, only R_q-module maps commute with the algebra, so either express A as a structured (negacyclic-block) matrix over R_q, or keep the A-application in the coefficient domain with the trace/constant-term techniques of [18, 21]. This is the one place the lift is design work rather than transcription; it is also exactly the machinery those papers supply.

Sublinear proofs. The linear-in-Nr proof sizes here are the entry point, not the ceiling: recursive folding systems for exactly this commitment family — LaBRADOR [22] and the Greyhound polynomial commitment [23] — compress proofs of huge linear/bilinear systems to tens of kilobytes, and matrix multiplication embeds naturally. A production system proving many products should prove them all inside one folded instance.

Alternatives worth pricing. If interaction or a transparent hash-based stack is acceptable and hiding requirements are mild, the sum-check protocol gives matrix-multiplication verification with prover overhead close to the matmul itself [24], at the cost of different (non-lattice, non-hiding by default) machinery. Conversely, if the deployment already runs an elliptic-curve stack, Pedersen-commitment versions of Π_1/Π_2 are the classical originals of these protocols — smaller proofs, no post-quantum claim.

Multi-prover splits. When A and B belong to *different* owners, Π_2 composes with secret-shared witnesses along the lines of collaborative proof systems [27]; the commit–compress–mask structure survives because every prover-side operation is linear until the single product gate.

Suggested build order: Route Z at set B → full test battery of Section 16 → estimator sign-off → Route R port with Z as differential oracle → folding layer only if proof size demands it.

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